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Dynamics of mesoscale brain network during visual discrimination learning revealed by chronic, large-scale single-unit recording

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eLife Assessment

This study presents experiments suggesting intriguing mesoscale reorganization of functional connectivity across distributed cortical and subcortical circuits during learning. The approach is technically impressive and the results are potentially of **valuable** significance. The authors have also made clear effort to address concerns in revision. However, the strength of evidence remains **incomplete**. Acquisition of data from additional animals in the primary experiment could bolster these findings.

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Abstract

Associating unfamiliar stimuli with appropriate behavior through experience is crucial for survival. While task-relevant information was found to be distributed across multiple brain regions, how regional nodes in this distributed network reorganize their functional interactions throughout learning remains to be elucidated. Here, we performed chronic, large-scale single-unit recording across 10 cortical and subcortical regions using ultra-flexible microelectrode arrays in mice performing a visual decision-making task and tracked mesoscale functional network dynamics throughout learning. Task learning reshaped interregional functional connectivity, leading to the emergence of a subnetwork involving visual and frontal regions during the acquisition of correct No-Go responses. This reorganization was accompanied by a more widespread representation of visual stimulus across regions, and a region's network rank strongly predicted its peak timing of visual information encoding. Together, our findings revealed that mesoscale networks undergo dynamic restructuring during learning, with functional connectivity ranks influencing the propagation of sensory information across the network.

Introduction

Learning to make appropriate decisions based on external stimuli is fundamental for survival and adaptive behavior. This process relies on the coordinated activity of distributed neural circuits spanning sensory, association, and motor areas. Previous studies have implicated multiple cortical and subcortical regions in visual task learning and decision-making ^{1–9}. In the cortex, anterior regions have been found to play important roles. The medial prefrontal cortex (mPFC) was indispensable for both learning visual discrimination and maintaining enhanced visual acuity after learning ². The secondary motor cortex (M2) encoded sensory history information in a flexible visual decision task, and its inactivation impaired adaptive action selection ³. Visuomotor learning also promoted visually evoked activity in M2 and anterior cingulate cortex (ACC) ⁴. Top-down inputs to the primary visual cortex (V1) are also critical. Orbitofrontal cortex (OFC) projections to V1 were required for learning a visual Go/No-Go task ⁵, and retrosplenial inputs to

V1 were essential for encoding task-related events⁶. Among subcortical regions, the dorsomedial striatum was necessary for visual category learning⁷, and innervation from M2 to the dorsal striatum suppressed inappropriate visual decisions⁸. The mediodorsal thalamus (MDTh) regulated prefrontal signal and noise via distinct circuit mechanisms under different scenarios of decision uncertainty⁹. Although the roles of these brain regions have been well studied individually or in pairwise combinations, how these regions dynamically reorganize their functional interactions as a mesoscale network during the learning of decision-making remains unclear. Addressing this question requires an approach that captures large-scale, longitudinal activity patterns across both cortical and subcortical areas.

Recent advances in large-scale neural recordings have enabled monitoring of activity across multiple brain regions, providing new insights into information representation and transformation at a mesoscale level^{10,11}. It has been revealed that the encoding of sensory, choice, and body motion information is not confined to a single or a few brain regions, but widely distributed across brain regions during visual decision tasks^{12–14}. Similarly, sensorimotor transformations during decision-making have been found highly distributed across many brain regions following the learning of a visual change detection task¹⁵. In a Go/No Go tactile discrimination task, multi-fiber photometry revealed functional networks encompassing basal ganglia, thalamus, neocortex and hippocampus grow and stabilize upon learning, and during learning most regions shift their peak activity from the time of reward-related action to the reward-predicting stimulus¹⁶. In a delayed-response paradigm, learning has been associated with emergence of a specific subnetwork involving layer 2/3 neurons in the anterior lateral motor cortex and posterior parietal cortex, accompanied by sparser global functional connectivity across the dorsal cortex¹⁷. However, most of these studies have focused on neural dynamics in expert animals or have been restricted to superficial cortical layers, largely due to technical constraints of commonly used techniques such as high-density silicon probes and calcium imaging. These approaches typically provide either limited spatial coverage in depth or lack the longitudinal recording capability across learning. As a result, how cortical-subcortical neural spiking dynamics evolve during task acquisition across broad spatial scales remains poorly understood.

To tackle this question, we utilized uFINE-M (ultra-Flexible Implantable Neural Electrodes for Mouse) arrays¹⁸ to simultaneously record spiking activity across 10 brain regions in mice learning a visual Go/No-Go task over two to three weeks. The chronic implantation capability of uFINE-M arrays enabled us to track the evolution of a mesoscale functional network including frontal regions (mPFC, OFC, and ACC), motor cortices (M1 and M2), visual cortices (V1, V2M, and V2L), and subcortical regions (striatum and MD thalamus) throughout learning. By analyzing functional connectivity patterns and information encoding dynamics, we found that learning reshaped interregional communication and accelerated the broadcast of stimulus information throughout the network. These findings provide insights into how distributed brain networks adapt during the acquisition of decision-making skills.

Results

High-throughput recording in mice performing a visual Go/No-Go task

To investigate the dynamics of the mesoscale functional network during decision-making task learning, we trained head-fixed mice to discriminate between two visual stimuli (vertical vs. horizontal static gratings) using a Go/No-Go paradigm, which has been used to study local neural dynamics during visual associative learning and decision-making⁵. In this task, mice were required to lick a waterspout in response to Go stimuli (Hit trials) within a specified response window to receive a water reward, while withholding licking for the No-Go stimuli (Correct rejection trials, CR trials) to avoid timeout punishment (Figure 1A [↗](#)). Despite substantial daily fluctuations in task performance (Figure S2 [↗](#)), mice generally gained proficiency over time by learning to make more correct rejection decisions at No-Go stimuli (Figure 1B [↗](#)).

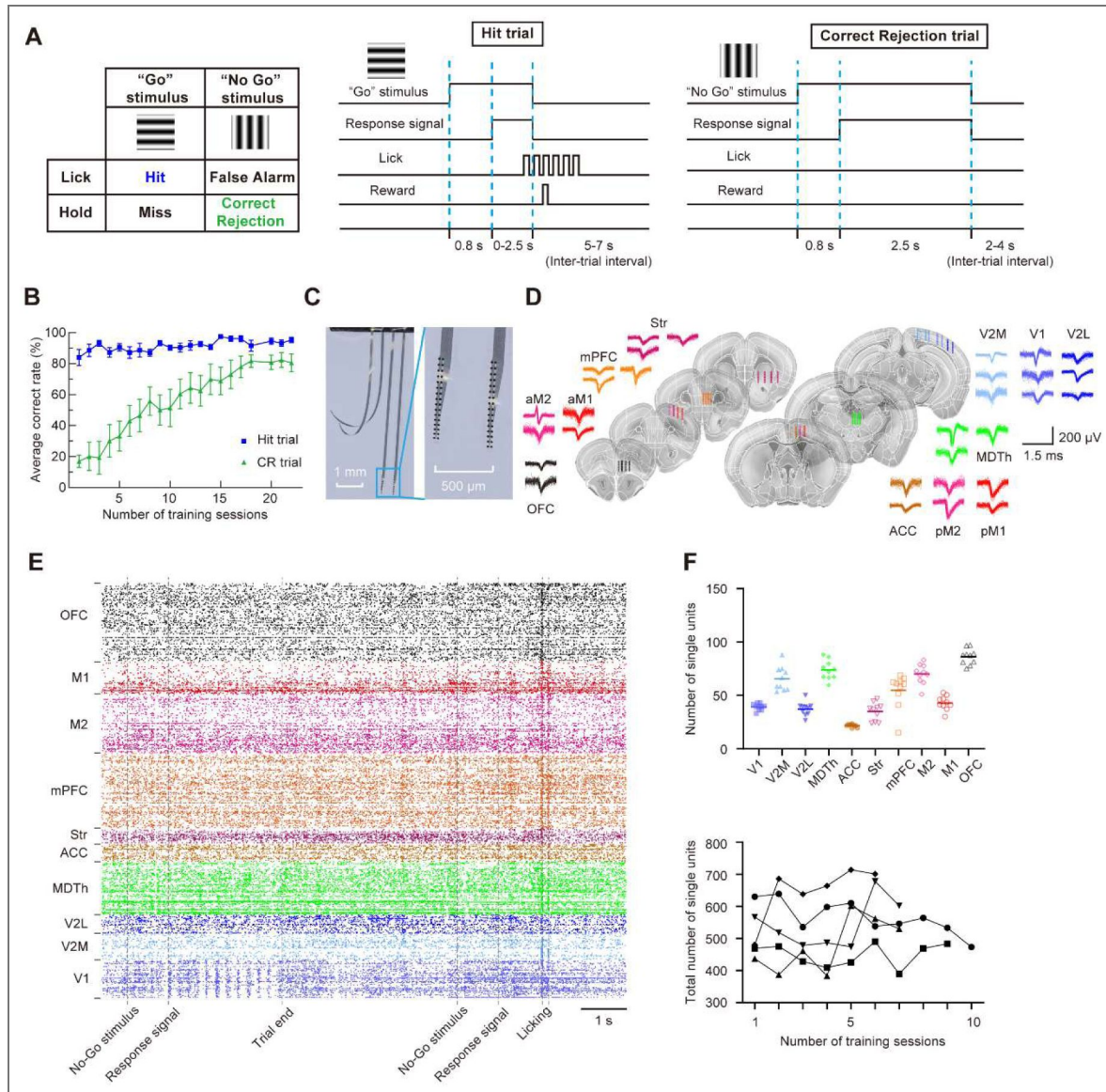


Figure 1. High-throughput recording in mice performing a visual Go/No-Go task.

(A) Schematic of the task. (B) Average correct rate during training (mean $\pm$ SEM, $n = 7$ mice). (C) Photos showing the uFINE-M shanks and recording sites. (D) Schematic showing the implantation sites of uFINE-M arrays, along with example single-unit waveforms recorded from each brain region. Brain section images are adapted from Allen reference atlas²⁵. (E) Example spike rasters during two trials. (F) Top, the number of single units recorded in each brain region. Each data point represents data from an individual recording session. Bottom, the total number of single units recorded during training ($n = 5$ mice). Each symbol represents data from an individual mouse.

To capture neural spiking activity across multiple brain regions throughout task learning, we chronically implanted eight 128-channel uFINE-M arrays into the left hemisphere of each mouse brain and simultaneously recorded from 10 brain regions, including frontal regions (mPFC, OFC, and ACC), motor cortices (M1 and M2), visual cortices (V1, V2M, and V2L), and subcortical regions (striatum and MD thalamus) (Figures 1C and D [↗](#)). These regions have been implicated in visuomotor tasks [3,5,7,8,19–21](#) and exhibit dense structural connectivity [22](#). The ultra-flexible arrays were guided to place by tungsten wires. We verified the accuracy of implantation depth of this implantation approach in a separate test group of mice (Figure S1 [↗](#)), though we observed a tendency for the arrays to end deeper than expected for cortex regions ($142.1 \pm 55.2 \mu\text{m}$, $n = 7$ shanks) and shallower for subcortical structures ($-122.6 \pm 71.7 \mu\text{m}$, $n = 7$ shanks), the extent of depth error was slight. On average, 532.1 ± 92.5 single units (mean $\pm$ SD, $n = 39$ sessions from 5 mice) were recorded across 1024 channels in each session, with no fewer than 15 units recorded from each region of interest (Figures 1E and F [↗](#)). Given the fluctuations in behavioral performance (Figure S2 [↗](#)), we categorized trials in early sessions with low behavioral discriminability [23](#) ($d\text{-prime} < 2$) as “early stage” data, and trials in late sessions with high behavioral discriminability ($d\text{-prime} > 3$) as “expert stage” data (Figure S2 [↗](#)). Only these data were used in the following analyses.

The average firing rate patterns showed substantial changes during learning (Figure 2A [↗](#)), with an overall decrease in firing rate during CR trials and increase in firing rate during Hit trials. Observing the low number of CR trials in early sessions resulted in noisier traces of firing rate, we accounted this by reconstructing bootstrap-resampled datasets with only 5 trials for each session in both the early stage and the expert stage. The mean trace of reconstructed datasets again showed overall decrease in CR trial firing rate during task learning (Figure S3 [↗](#)). Moreover, many brain regions exhibited significant changes in their temporal profile of activity in CR trials as learning progressed (Figure 2A [↗](#), Figure S3 [↗](#)).

Based on these observations, we calculated the activity onset timing of each single unit relative to the visual stimulus onset (Methods) and found that the mean activity onset timing formed a clearer sequential activation pattern across brain regions in expert CR trials (Figures 2B and C [↗](#)). This sequential pattern was accompanied by a more temporally compressed activation profile with learning, as evidenced by a significant reduction in the spread of peak activation times across regions (Figure S4 [↗](#) and Methods, early: 170.0 ± 160.0 ms vs. expert: 56.67 ± 38.58 ms, mean $\pm$ SD, $p < 1.0 \times 10^{-4}$, t-test with Sidak correction). Additionally, several visual and frontal regions (V1, V2M, mPFC, OFC, M2, and M1) also showed faster average response time following stimulus onset in expert CR trials compared to the early stage (Figure 2D [↗](#)). This learning-induced compression of mesoscale activity is reminiscent of similar phenomena observed in motor skill learning [24](#), suggesting a potential general principle of temporal refinement in distributed brain networks during task acquisition.

In contrast, learning this visual Go/No-Go task did not induce significant changes in the average activity onset timing for most regions in Hit trials (except V2L, Figure 2D [↗](#)), nor did it significantly alter the spread of regional peak activation times (Figure S4 [↗](#)). This could be attributed to the fact that the Go stimulus was already introduced during pre-training stages, when the mice learned the basic trial structure (Methods).

In summary, we found that learning this visual Go/No-Go task led to a sequential activation pattern of the neural activity across brain regions, particularly generating an earlier and more compressed activity sequence in CR trials. These results led us to further explore the leading-following relationships between regions and the roles of individual brain regions within the functional brain network. We focused specifically on CR trials, as mice improved their performance mainly by learning to correctly reject No-Go stimuli.

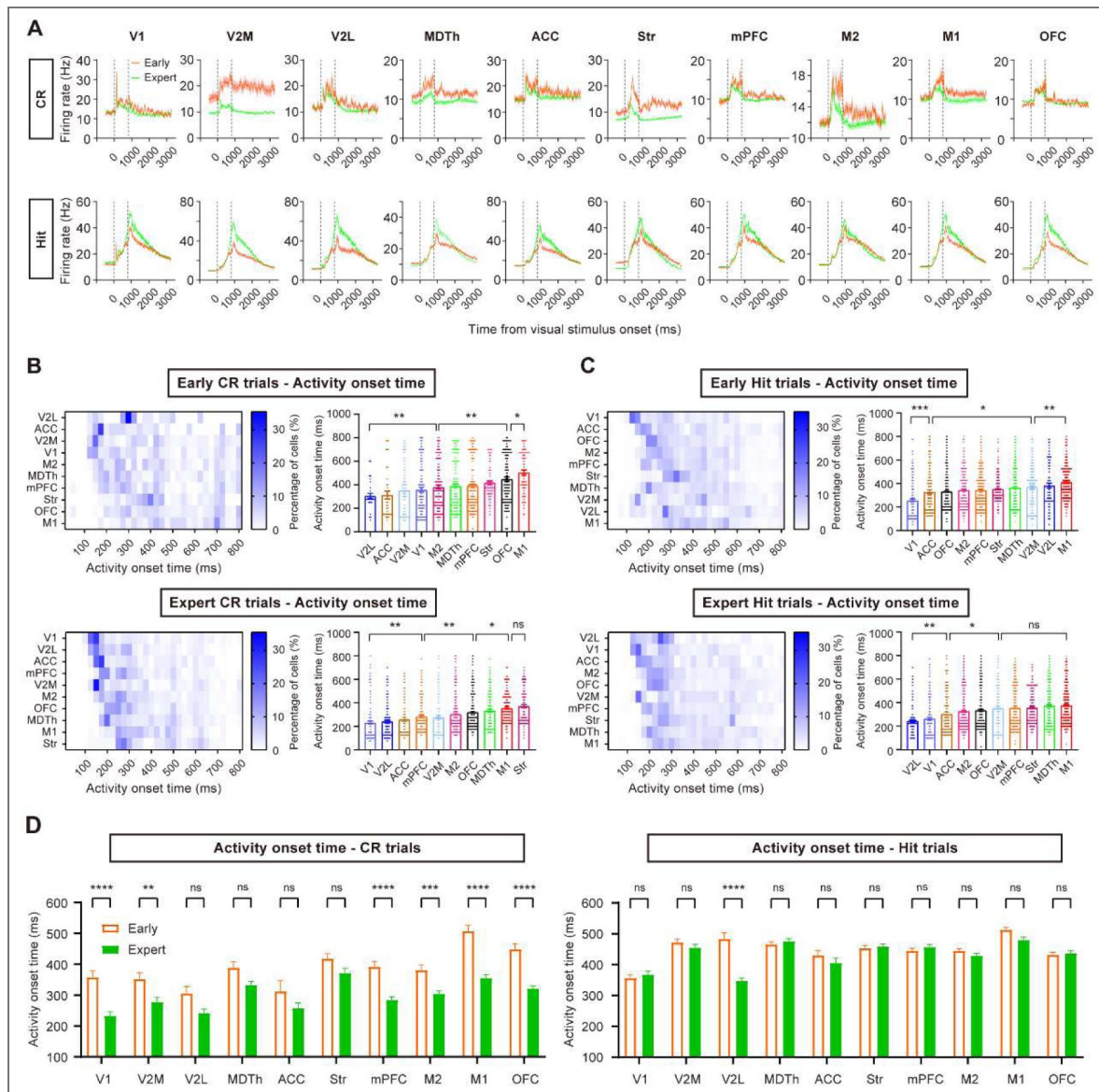


Figure 2. Activity changes throughout task learning.

(A) Averaged firing rate aligned to the visual stimulus onset for all CR trials and Hit trials in the early and expert stages ($n = 118$ early CR and 828 early Hit trials from 7 sessions of 3 mice, 610 expert CR trials and 677 expert Hit trials from the same mice). Shading, SEM. (B) Left, distribution of activity onset timing across time. Right, activity onset timing of each region. Each data point represents data from a neuron. (C) Same as B but for Hit trials. (D) Comparison of activity onset timing between the early and expert stages. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$, t-test with Sidak correction. Error bars, SEM.

Ranking dynamics of mesoscale brain network during learning of CR trials

To study functional connectivity dynamics and quantify the overall extent of leading-following relationship among spiking activity across brain regions, we first identified neuron pairs that exhibited functional connectivity if their cross-correlation score of spiking activity (TSPE algorithm)²⁶ was above chance level ($p < 0.05$, Figure 3). We focused on fast connections within 20 ms and included only excitatory connections in subsequent analyses. If the spiking activity of neuron *a* preceded that of neuron *b*, we defined neuron *a* as having functional output to neuron *b*, and neuron *b* receiving functional input from neuron *a*. The functional input/output strength between any two brain regions was then defined as the proportion of neuron pairs with significant excitatory functional input/output, relative to the total number of possible input/output neuron pairs between these two regions. Considering that differences in firing rates might bias cross-correlation between spike trains²⁷, making raw counts of significant neuron pairs difficult to compare across conditions, we ranked the values in the regional connection matrix on a scale from 1 to 10. This ranking approach enabled us to focus on the relative importance of each region within the brain network and more effectively evaluate the ranking dynamics across time windows and trial types.

In early CR trials, most brain regions did not show obvious differences in input/output rankings (Figure 4A), with rank values remaining close to 5 (the expected level for random data) throughout the CR trial. Only the ACC and V2L maintained high rank values in early CR trials, suggesting their roles in visual attention and high-level visual processing. After the mice achieved proficiency in this task, we observed a clear separation of input/output rankings among different brain regions (Figure 4A). ACC maintained a high rank, whereas the ranks of V2L, striatum, and MDTh decreased across all trial periods. In contrast, the ranks of V1, V2M, and OFC increased across all trial periods, and M2 exhibited an increased rank during the response period (Figures 4B, Figure S5, $p < 0.05$, t-test with Sidak correction). These results suggest that, during the learning of visual-based decision-making, brain regions within the network differentiate in task involvement. Regions associated with visual processing, value processing, and action selection can emerge as key input/output hubs, forming a more task-relevant subnetwork.

Since previous studies have reported the dominance of movement-related activity across brain regions^{13,28}, we examined the extent to which the observed changes in functional connectivity patterns could be explained by potential changes in body movements during task training. We performed video recording in a cohort of mice and quantified motion energy of facial movements, foot movements, and pupil dynamics²⁹ during CR trials. Motion energy for face and pupil showed a reduction across all CR trial time windows during learning (Figure S6), which might account for the overall decrease in firing rates in CR trials (Figure 2A), and the decrease for functional connection rank of the striatum. Only feet movements showed a brief increase during the stimulus period (Figure S6). Thus, the changes in body movements could not fully explain the stable separation of functional connection ranks in the stimulus and responses periods of CR trials during learning.

Ranking dynamics during learning of Hit trials and fruitless learning

Compared to CR trials, the mesoscale network showed more rapid and dynamic transitions at different intra-trial time points in Hit trials (Figure 5A). In early Hit trials, a transient separation of input/output rankings was observed around the visual stimulus onset, with the ACC ranking the highest, which is similar to CR trials. During the response period, regional rankings become more convergent. After the mice reached expert level in this task, although the correct rate of Hit trials remained unchanged (Figure 1B), the striatum acquired a high input ranking, particularly during the early response period of expert Hit trials (Figures 5A and B, Figure S7A). The rise of input rank of striatum during the response period was still clear with the analyses repeated on spike time data aligned to the first lick in each Hit trial, indicating this observation was not a result

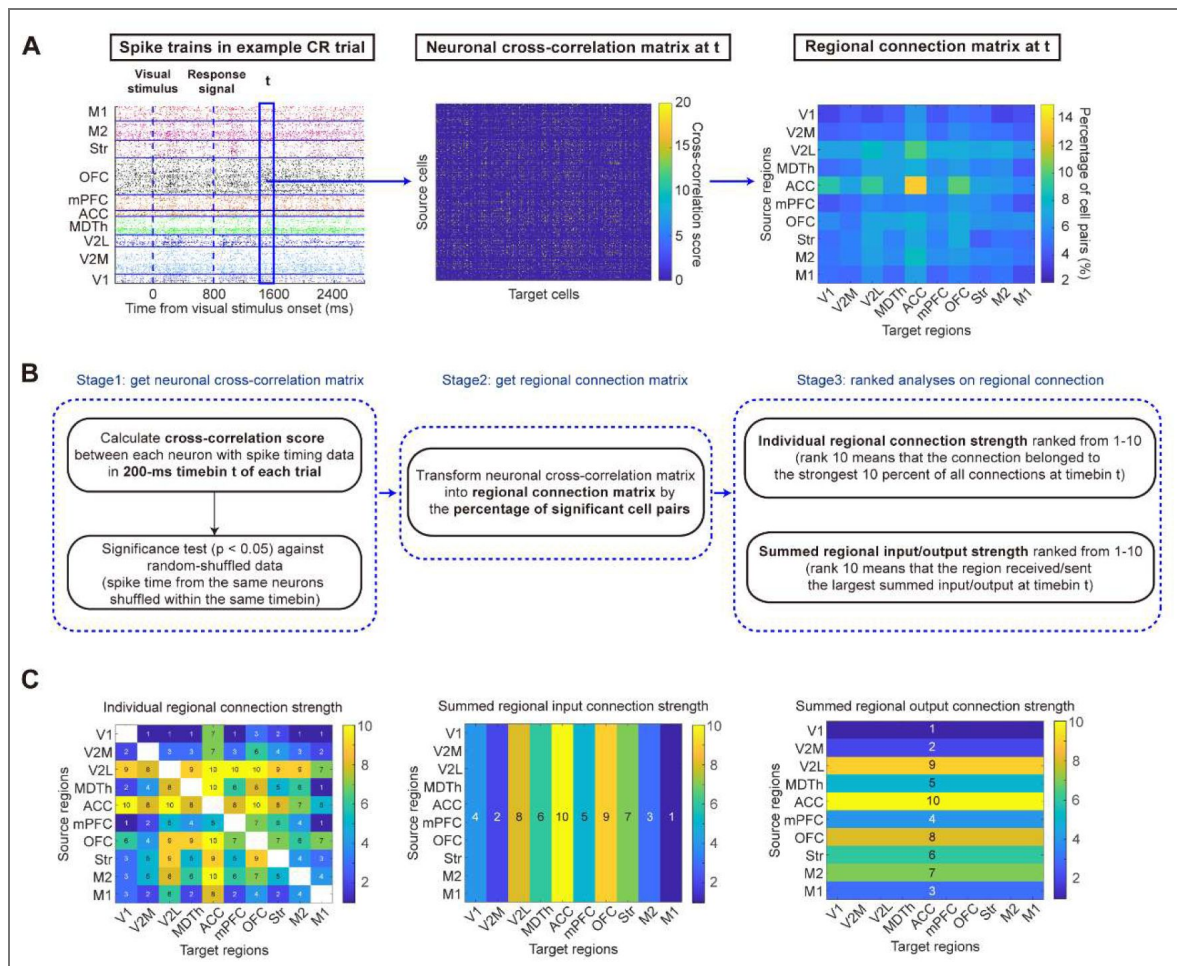


Figure 3. Definition of functional connection.

(A) Schematic of data processing flow of calculating functional connectivity. For each 200-ms time window (t), cross-correlation scores were calculated between spike trains of neuron pairs and the percentage of neuron pairs that showed significant cross-correlations was treated as functional connection strength between brain regions. The regional connection matrix was then ranked from 1 to 10 to evaluate the relative importance of regional connection compared with other connections within the same time window of the same trial. (B) Details of processing stages. (C) Connection rank matrix calculated from the example data in A. The individual regional connection strength measurement (left) was used in Figure 6 and Figure S10. The summed regional connection strength measurement (middle and right) was used in Figure 3-5 and 7, Figure S5 and S5-7.

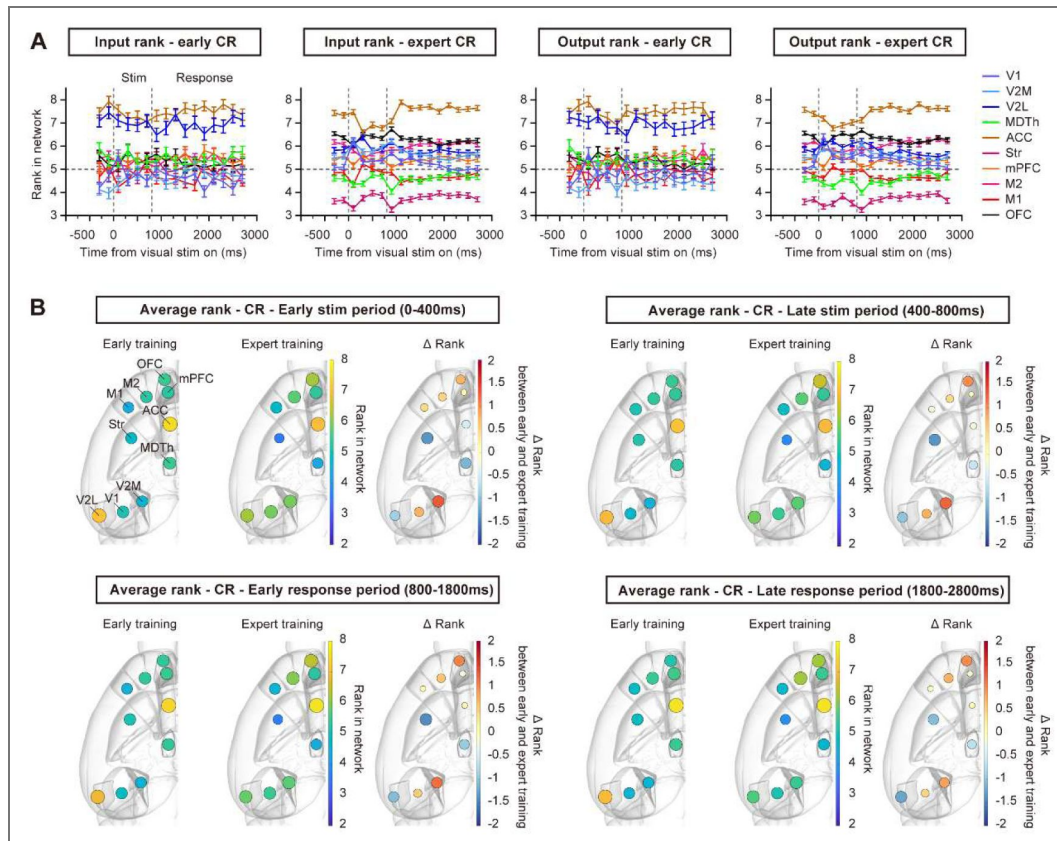


Figure 4. Ranking dynamics in CR trials during learning.

(A) Input/output ranking dynamics during early and expert CR trials. (B) Average rank of each brain region in the early stimulus period (0–400 ms after stimulus onset), late stimulus period (400–800 ms after stimulus onset), early response period (800–1800 ms after stimulus onset), and late response period (1800–2800 ms after stimulus onset) of early and expert CR trials, mapped on brain atlas³⁰. Δ Rank represents the rank change between the expert and early stages. $n = 118$ early CR trials from 7 sessions of 3 mice, and 610 expert CR trials from 6 sessions of same mice. Error bars, SEM.

of possible changes in lick initiation time during learning (Figure S8 [↗](#)). Still, we cannot fully rule out the effects from more subtle movement changes during learning, since the motion energy for face also increased in early response period (Figure S6 [↗](#)). In addition to the striatum, we also observed significant changes in the input/output ranks of multiple regions between early and expert Hit trials (Figure S7 [↗](#)), suggesting that distinct mesoscale functional connectivity patterns could underlie similar behaviors.

We also trained mice on a “fruitless learning” task, in which all visual stimuli and task structure were identical to those in the normal learning group, but the Go/No-Go visual stimuli were presented randomly and had no association with reward. As expected, mice continued to lick in every trial regardless of the visual stimulus type, and we defined the trials in which mice were randomly rewarded as fruitless-learning Hit trials. During the early stimulus period, the ACC and visual regions showed the highest ranks in these fruitless-learning Hit trials (Figure S9 [↗](#)), similar to those in the normal learning group. In the early response period, however, we observed a strong elevation of the rank of the MDTh (Figure S9 [↗](#)), suggesting cognitive effort in the face of uncertainty regarding the task rules [31,32](#). These results further demonstrate that distinct mesoscale functional connectivity patterns can emerge and evolve depending on task demands, in accordance with previous reports [17,33–35](#).

Rank increase of the visual/frontal regions was attributed to elevated regional connection rank in CR trials

To investigate the factors driving the rank changes in CR trials during learning, we examined the regional connection ranks of brain regions that showed rank increases in CR trials (V1, V2M, M2, and OFC, Figures 4B and C [↗](#), Figure S5 [↗](#)). For each time window within a trial, regional connection strength was ranked on a scale from 1 to 10, with a rank of 1 representing the lowest 10% strength among all regional connections within the same time window (Figures 3B and C [↗](#)). We observed a general increase in the ranks of regional connections between these regions (Figure 6 [↗](#), V1, V2M, M2, and OFC) and other frontal and motor regions (Figure 6 [↗](#), mPFC, ACC, M2, and M1).

We also examined the regional connection ranks of regions that exhibited rank decreases in CR trials (V2L, MDTh, and striatum, Figure S10 [↗](#)). All three regions showed a decline in regional connection rank both with each other and with most frontal and motor regions (mPFC, ACC, M2, and M1). The striatum, which exhibited the most pronounced rank decrease, showed the most widespread reduction in regional connection ranks.

In summary, these results suggest that the network forms a more compact functional subnetwork during learning to reject the No-Go visual stimulus. This reorganization is characterized by increased relative connection strength among several key visual (V1 and V2M), frontal (mPFC, OFC, and ACC), and motor regions (M2 and M1), while regions such as V2L, MDTh, and the striatum become less engaged in the task-related functional network.

Visual stimulus information became widespread in the stimulus period as learning progressed

After examining network dynamics during different trial periods and learning stages, we wondered how the stimulus encoding ability of each region changed during task learning. To assess the stimulus encoding ability based on spike counts, we grouped trials according to visual stimulus identity (with behavioral choice balanced) and applied ROC (receiver operating characteristic) analyses in each 200-ms time window [23](#). In each session, spike count data for each neuron was bootstrap-resampled to balance the number of trials across different trial types (Figure 7A [↗](#)). For each neuron, 50 trials were resampled with replacement for each trial type to perform ROC analyses, and this procedure was repeated 500 times for each time window. A neuron was classified as stimulus-selective if its ROC selectivity was above 95% of its own randomly shuffled spike count data ($p < 0.05$) in more than 95% of resampling iterations.

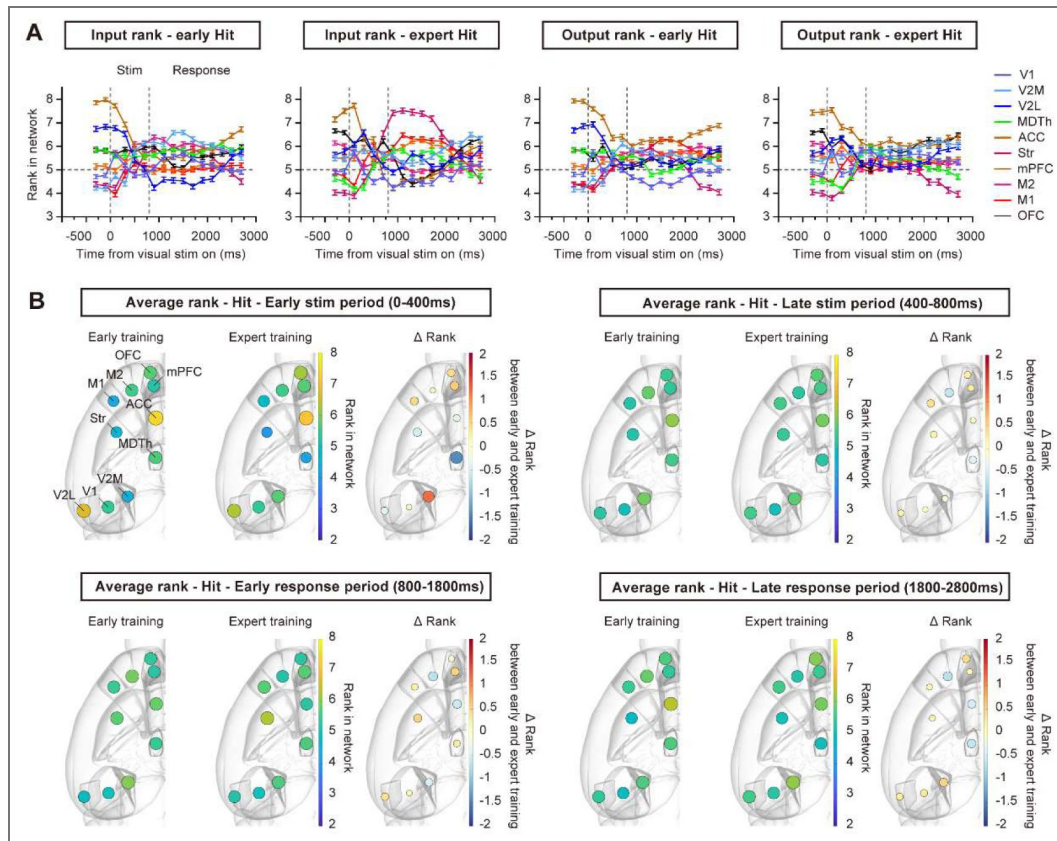


Figure 5. Ranking dynamics in Hit trials during learning.

(A) Input/output ranking dynamics in early and expert Hit trials. (B) Average rank of each brain region in the early stimulus period (0–400 ms after stimulus onset), late stimulus period (400–800 ms after stimulus onset), early response period (800–1800 ms after stimulus onset), and late response period (1800–2800 ms after stimulus onset) of early and expert Hit trials, mapped on brain atlas³⁰. ΔRank represents the rank change between the expert and early stages. $n = 828$ early Hit trials from 7 sessions of 3 mice, and 677 expert Hit trials from 6 sessions of 3 mice. Error bars, SEM.

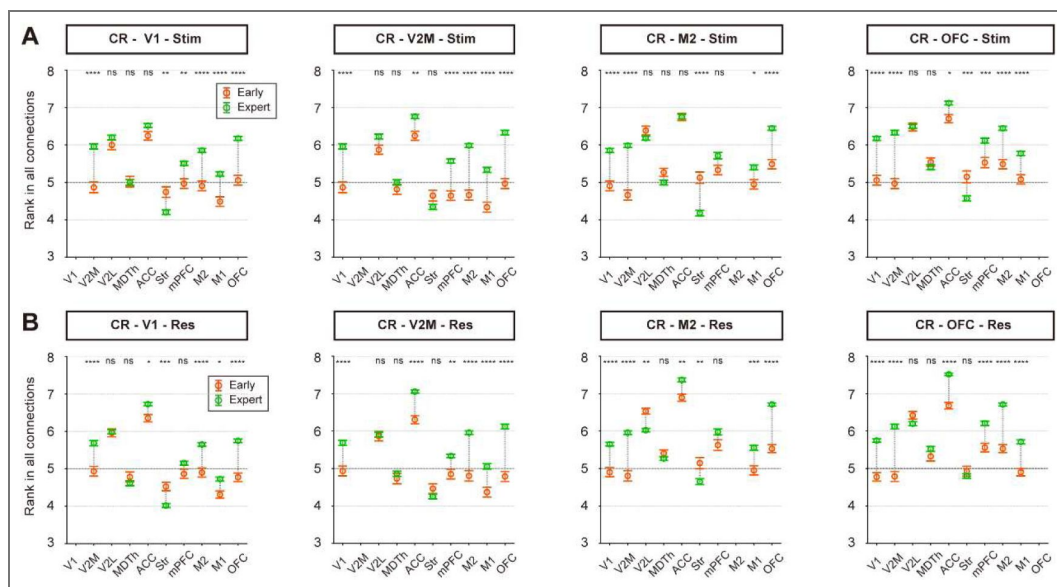


Figure 6. Rank increase in CR trials was attributed to elevated input/output rank from/to other regions.

(A) Average functional connection rank changes for the four regions (V1, V2M, M2, and OFC) that showed increased rank values in the stimulus period of CR trials during visual learning. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$, t-test with Sidak correction. (B) Same as A but for the response period of CR trials. $n = 118$ early CR trials from 7 sessions of 3 mice, and 610 expert CR trials from 6 sessions of 3 mice. Dashed lines at rank 5 indicate the average level of random data. Error bars, SEM. Stim, stimulus period. Res, response period.

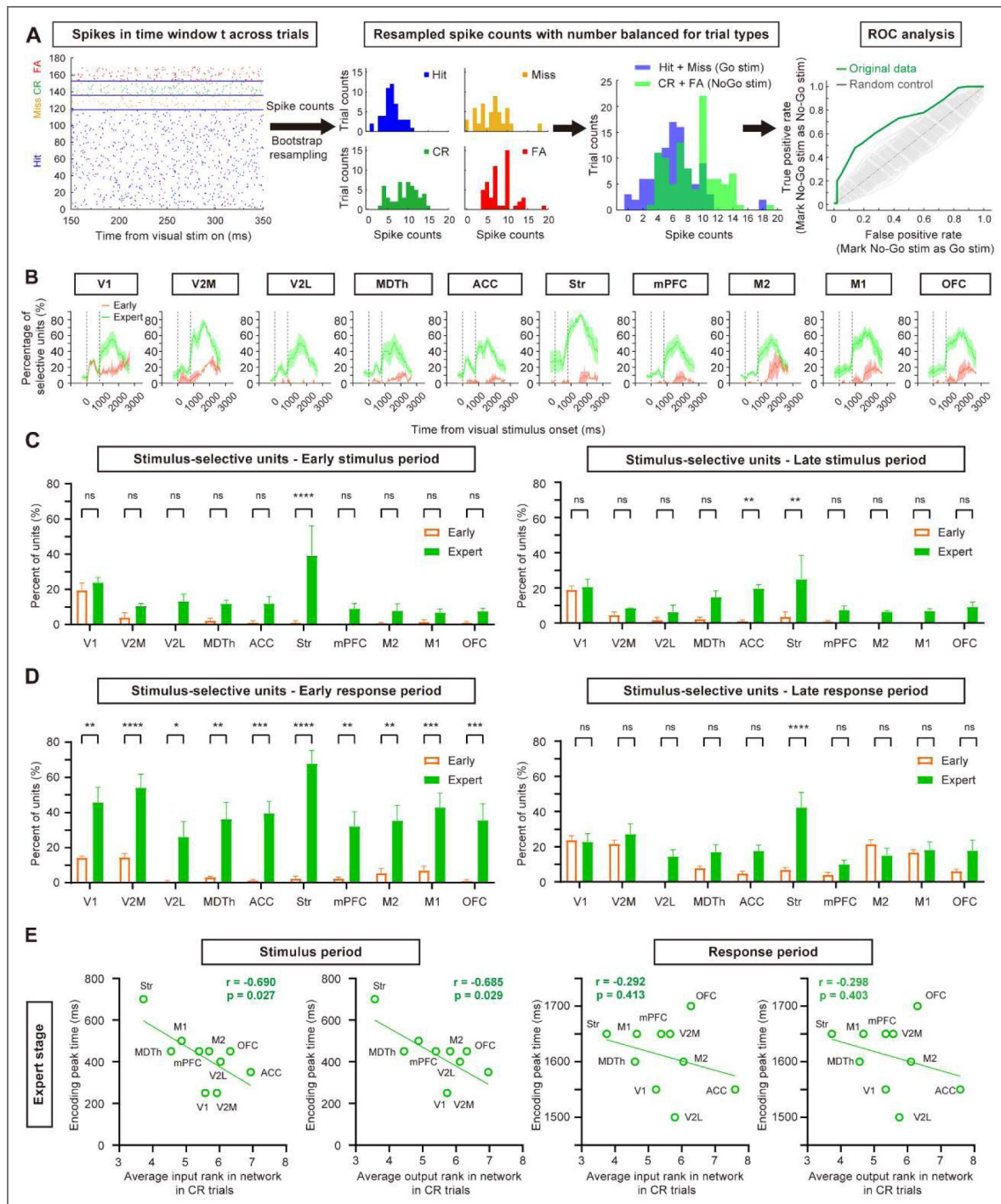


Figure 7. Encoding of visual stimulus information during task learning.

(A) Schematic of the ROC analyses and example data from a neuron preferring the No-Go stimulus. (B) Percentage of stimulus-selective neurons in each brain region during the early and late training stages. (C) Mean percentage of stimulus-selective neurons in the early (0–400 ms after stimulus onset) and late (400–800 ms after stimulus onset) stimulus periods. $n = 10$ time bins for the early training stage and 30 time bins for the expert training stage. (D) Same as C, but for the early (800–1800 ms after stimulus onset), and late (1800–2800 ms after stimulus onset) response periods. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$, t-test with Sidak correction. $n = 34$ time bins for the early training stage and 102 time bins for late training stage. (E) Correlation between stimulus encoding peak time and input/output rank in expert CR trials. A significant correlation was observed during the stimulus period but not the response period (Pearson's correlation).

In the early training stage, stimulus-selective neurons were mainly found in V1 during the stimulus period (Figures 7B and C), while other regions contained very few stimulus-selective neurons during this period. By the late response period, stimulus-selective neurons were found in larger proportions in nearly all regions (except V2L), suggesting that the visual information was broadcast through the network at this time (Figures 7B and D). In the expert stage, however, stimulus-selective neurons emerged in all regions during the stimulus period, and their proportion also substantially increased during the response period (Figure 7D), in alignment with previous reports that visuomotor learning could promote visually evoked activity in dorsal medial prefrontal cortex⁴, though this could also be the result from the potential movement difference in FA trials and Hit trials (Figure S6). We also noticed that some regions showed stimulus encoding even before the visual stimulus onset, suggesting effects from trial history³⁶ or expert mice had likely learned the pseudo-random trial sequence (Methods) and anticipated upcoming visual stimuli based on sensory history.

In expert mice, ROC encoding curves of most regions showed two distinct peaks—one during the stimulus period and another during the response period (Figure 7B, except the striatum curve, which ramped up in stimulus period and only showed one peak in response period). Therefore, we defined the peak time of stimulus encoding in each trial period as the center of the time window with the highest mean percentage of stimulus-selective neurons. We found a significant correlation between the input/output rank of each brain region in CR expert trials and its encoding peak time during the stimulus period ($p < 0.05$, Pearson's correlation, Figure 7E), with higher-ranked regions reaching their encoding peaks earlier. No significant correlation was observed during the response period in expert CR trials (Figure 7E).

In summary, as learning progressed, visual information propagated more rapidly through the network, likely due to the increased functional connection ranks between visual and frontal regions. Moreover, a region's connection rank within the network became highly predictive of how quickly it reached its encoding peak during stimulus viewing.

Optogenetic inhibition of rank-increasing regions impaired task learning

Finally, to examine whether the regions with increased rank during CR trials actually contributed to task learning, we performed manipulation experiments on two of these regions, specifically V2M and OFC. For each manipulation group, we expressed AAV2/9-mCaMKIIa-eJaws3.0-mRuby3-WPRE-pA (AAV2/9-mCaMKIIa-mCherry-WPRE-pA for the control group) in the bilateral OFC or V2M and inhibited these regions during either the stimulus or response period of task training (Figure 8).

We found bilateral inhibition of the OFC (Figures 8B and C) showed significantly impaired task learning in both the stimulus period ($n = 8$ mice for eJaws 3.0 group, $n = 14$ mice for mCherry group, $F(1, 399) = 29.91$, $p < 1.0 \times 10^{-4}$, $\eta^2 = 0.047$, two-way ANOVA), and the response period ($n = 8$ mice for eJaws 3.0 group, $n = 16$ mice for mCherry group, $F(1, 420) = 87.51$, $p < 1.0 \times 10^{-4}$, $\eta^2 = 0.098$, two-way ANOVA).

The interaction with training sessions was not significant for both periods ($F(19, 399) = 0.75$, $p = 0.77$, $\eta^2 = 0.022$ for the stimulus period, $F(19, 420) = 1.05$, $p = 0.40$, $\eta^2 = 0.022$ for the response period), suggesting consistent impairment across training sessions. Bilateral inhibition of V2M during the stimulus period also impaired task learning with a small effect size ($n = 8$ mice for eJaws 3.0 group, $n = 14$ mice for mCherry group, $F(1, 399) = 8.19$, $p = 4.4 \times 10^{-3}$, $\eta^2 = 0.012$, two-way ANOVA, Figure 8F), whereas inhibition during the response period did not affect task learning ($n = 8$ mice for eJaws 3.0 group, $n = 16$ mice for mCherry group, $F(1, 440) = 0.0095$, $p = 0.92$, two-way ANOVA, Figure 8G). None of the manipulation groups showed significant differences in miss rate compared to the mCherry control group ($p > 0.05$, Welch's *t* test, Figures 8D and H), indicating the observed performance decline was not due to task abandonment.

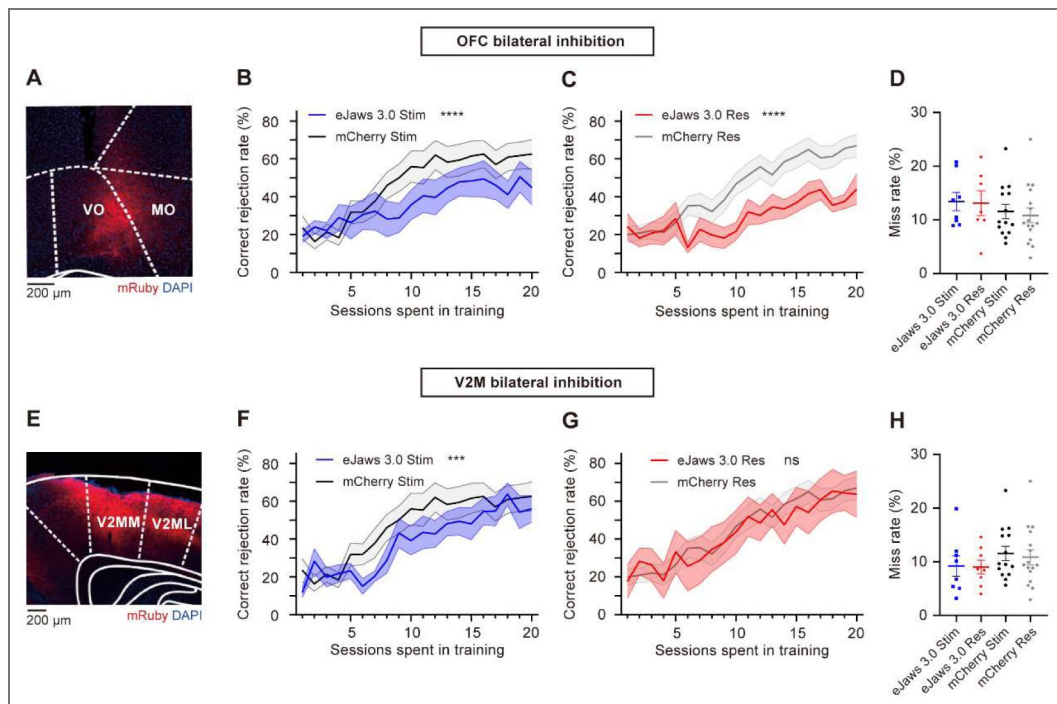


Figure 8. The effects of bilateral optogenetic inhibition on task performance.

(A) Expression of AAV2/9-mCaMKIIa-eJaws3.0-mRuby3-WPRE-pA in the OFC. VO: ventral orbitofrontal cortex; MO: medial orbitofrontal cortex. Regions were named according to the Paxinos atlas³⁷. (B) Correct rejection rate for the OFC-stimulus period inhibition group (eJaws 3.0 Stim), and the control group (mCherry Stim). Shading, SEM. $n = 8$ and 14 mice for the eJaws 3.0 and mCherry group, respectively. **** $p < 0.0001$, significance for the group factor in two-way ANOVA. (C) Same as B, but for the OFC-response period inhibition group ($n = 8$ mice) and control group ($n = 16$ mice). (D) Average miss rate for each mouse in the OFC manipulation group and control group. (E-H): Same as A to D but for V2M inhibition. $n = 8$ mice for each manipulation group. *** $p < 0.001$, significance for the group factor in two-way ANOVA. The control group here was the same group of mice in A-D.

Taken together, the manipulation effects on task performance provide some support for the connection rank analysis, suggesting that regions with increased rank during learning likely contribute to task acquisition. However, while a rise in connection rank may reflect a region's involvement in the learning process, it does not necessarily imply a causal relationship with learning.

Discussion

In this study, we investigated how mesoscale functional networks changed during the learning of a visual-based decision-making task. Using 1024-channel uFINE-M arrays for chronic spiking activity recording across multiple cortical and subcortical regions, we were able to examine the mesoscale network dynamics at different timescales: rapid transitions between different periods within a trial, distinct functional connectivity patterns across trial types within a session, and the long-term evolution of network dynamics throughout task learning.

A key finding of our study is that task learning reshaped interregional connectivity, leading to the emergence of a more task-relevant subnetwork as mice learned to correctly reject No-Go stimuli. Specifically, several visual and frontal regions (V1, V2M, OFC, and M2) gained prominence in the network, while others (V2L, MDTh, and striatum) became less engaged. These findings suggest that learning is accompanied by a selective refinement of interregional communication, with a shift in functional connectivity toward regions more directly involved in processing task-relevant information, and shifts in peak activity time to form a faster and more compressed activity sequence across regions. These observations align with previous reports of the dynamic reorganization of cortico-basal ganglia-thalamo-cortico network during the learning of a tactile discrimination task ¹⁶, the spatiotemporal refinement in cortical activity during the learning of a texture discrimination task ³⁸ and a visually guided delayed-response task ¹⁷, and the enhanced coupling between somatosensory neurons and frontal neurons in a whisker detection task ³⁹, suggesting that the emergence of a more task-relevant functional network may be a general mesoscale network feature of learning.

Beyond connectivity changes, we also found that the encoding of stimulus information became more widely distributed across the network as learning progressed. Moreover, the connectivity rank of a brain region was strongly correlated with the timing of its stimulus encoding peak during the stimulus period, suggesting that high-ranked regions may not only receive information earlier but also play a more central role in relaying task-relevant signals. These findings indicate that learning facilitates more efficient information flow through the network, potentially enhancing sensory processing and decision-making processes. The broader recruitment of stimulus-selective neurons during the response period in expert mice further supports the notion that learned associations between sensory inputs and behavioral outcomes become increasingly embedded in distributed circuits over time ^{14,15}.

We also noticed discrepancies between the results of network ranking analyses and optogenetic inhibition experiments. Inhibiting OFC during either the stimulus or response period significantly impaired learning, consistent with its increased rank in both task periods. However, while V2M also showed increased network rank in both the stimulus and response periods, inhibition of V2M during the response period had no significant effect. This suggests that rank increases with learning do not necessarily indicate a direct causal role in driving behavioral improvements. Other factors, such as neuromodulatory influences and internal state changes, may also contribute to the observed changes in functional connectivity ⁴⁰. We also cannot fully rule out the possible effect from rebound activity following optogenetic inhibition ^{41,42}, which may confound the interpretation of manipulation effects during the stimulus period (Figure 8 [↗](#)), but did not change the causal role of OFC in learning and the discrepancies between ranking analysis and the manipulation results in this case.

Despite recording from 10 brain regions, our study remains limited compared to the extensive network of brain regions implicated in visual-based decision-making ¹⁴, including the midbrain, hindbrain, and cerebellum. The limited anatomical coverage and relatively simple task design

might restrict the generalizability of our findings to more complex forms of decision making, and the observed changes in ranking dynamics could also arise from broader shifts in arousal, attention, or motivation over repeated sessions. We also didn't succeed in chronically tracking enough number of neurons for each region, which denied the chance to investigate single neuron level functional connection changes during learning. Considering that differences in firing rates might bias cross-correlation between spike trains²⁷ and making raw counts of significant neuron pairs difficult to compare across conditions, we focused on the relative importance of each region within the brain network and took a ranking approach to more effectively evaluate the ranking dynamics across time windows and trial types.

But this approach remained descriptive and might obscure magnitude of differences in connection strengths. Finally, though we balanced the number of each trial type in the encoding analyses, we cannot fully rule out the influence from differences in movements between Hit trials and FA trials, which might explain the large percentage of encoding neurons in late response period. There has been extensive literature^{12,13,28,43} on the strong effects of body movements on brain dynamics, and we also found the motion energy dynamics of the mice could explain the broad decrease in CR trial firing rates, and the decline in functional connection rank of the striatum. Though we found decrease in motion alone cannot fully explain the development of functional connection dynamics across the network, since the mice used in motion analyses were from a separate group, with only limited body parts monitored during learning, we could not further disentangle movement-related neural activity from task-related signals. Given these limitations, future studies should aim for broader spatial coverage, ideally with stable tracking of the same neuronal populations throughout the entire learning process, to achieve a more comprehensive characterization of the mesoscale network dynamics. Additionally, carefully designed decision-making tasks⁴⁴ will be essential for disentangling neural representations of sensory stimulus information, decision-making, action execution, and arousal states. It is also essential to establish more reliable analysis approaches that could provide more accurate assessment of functional connection dynamics without bias from firing rates. More importantly, as recording scales continue to expand, future work should aim to systematically evaluate the predictive power of different analysis methods in determining the causal contributions of various brain regions to learning and decision-making.

Materials and methods

Animals

Animal use procedures were approved by the Animal Care and Use Committee at the Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences (approval number NA-056-2023). Data were collected from a total of 87 male adult C57BL/6 mice (3–5 months old). Among them, 7 were used for task training to acquire behavioral learning results without electrode array or optical fiber implants, 5 for electrophysiological recordings during behavioral task (3 for visual-based decision-making learning and 2 for fruitless learning), 69 for optogenetic manipulation experiments, 3 for video analysis, and 3 for verification of implantation approach. Mice were generally housed in groups of 3–4 per cage, but mice for chronic extracellular recordings were housed individually to protect the implants. Mice were water-deprived in the home cage and received water reward during daily behavioral sessions. On days when mice did not perform the task, restricted water access (~0.8 mL per mouse) was provided each day. All mice were maintained on a 12-h light/12-h dark cycle (lights on at 7:00 a.m.), and all sessions were performed in the light phase.

Visual Go/No-Go task

Mice were head-fixed during training sessions and positioned in an acrylic tube placed in a behavioral chamber. A capacitive lick detector and a peristaltic water pump were controlled by custom MATLAB (MathWorks) scripts and digital I/O devices (Arduino Uno R3, Arduino) to monitor tongue licks and deliver water reward, respectively. Visual stimuli were presented on a 19" LCD

monitor (Dell P1917S, max luminance 80 cd/m²) placed 10-cm from the right eye of the head-fixed mouse. A yellow light-emitting diode (LED) was placed above the waterspout to signal the onset of the response period (response signal).

Each trial was initiated automatically after the preceding inter-trial interval (ITI) expired. A full-field visual stimulus (vertical or horizontal static gratings with spatial frequency of 0.09 cycles/° and 100% contrast) was presented on the monitor for 800 ms, followed by illumination of the yellow LED to indicate the start of the response window (response signal). Mice were required to lick during the response period of “Go” stimuli to receive a water reward (Hit trials). Failure to respond by licking during the response window of “Go” stimuli would result in a Miss trial with no reward or punishment. Licking during the response period of “No-Go” stimuli would be punished with an 8-s timeout (False Alarm trials, FA). Correctly withholding licking for “No-Go” stimuli (Correct response trials, CR) would be rewarded with a 2-s reduction in the ITI. ITIs were randomized between 4–6 s, but lick during the ITI would extend the interval by an additional 4–6 s (up to a maximum of 30 s) to punish impulsive licking. The training session was terminated if there were no lick responses in 20 consecutive trials.

The mice were trained to perform this task in three sequential steps. In step 1 (day 1–2), mice were allowed to collect water rewards by simply licking the waterspout placed under nose, with a fixed interval of 4 s. In step 2 (day 3–5), mice were required to lick specifically during the response window (at the presence of the LED response signal) and refrain from impulsive licking during the ITI. Only the “Go” stimulus was presented in step 2. In step 3 (day 6–20), the “No-Go” stimulus was introduced, and mice were trained to withhold licking for “No-Go” stimulus to avoid timeout punishment (Figure 1A). The trial sequence was pseudo-randomized to maintain a balanced number of “Go” and “No-Go” stimuli in every 6 trials.

For mice used in electrophysiological recordings, each mouse underwent 7.67 ± 2.08 (mean $\pm$ SD) sessions in training step 3 until they reached an average correct rate of $\sim 85\%$ in daily sessions. For optogenetics manipulation experiments, mice completed a fixed 20-session training in step 3.

Design, fabrication, and assembly of ultra-flexible microelectrode array (uFINE-M)

Each microelectrode array contained 128 channels for electrophysiological recording (Figure 1C). The array consisted of four flexible implantable shanks, each with 32 recording sites arranged in a 16×2 matrix. Each recording site was circular, with a diameter of 20 μm . The flexible shank was 6 mm in length, and the longitudinal spacing of electrode recording sites was either 30 μm or 50 μm to suit different spatial coverage requirements. The shank spacing was customized by adjusting the spacing between the four shuttling tungsten wires, which were used to guide the flexible shanks into brain tissue, ensuring proper alignment with the targeted implantation region (Figure 1D).

The fabrication of uFINE-M was adapted from a planar microfabrication technique featuring a multilayer architecture, as previously described¹⁸. The structural and passivation material was non-photosensitive polyimide, and patterning was achieved through O₂ plasma etching. Titanium was used as the adhesion layer between metal and polymer. The overall device thickness was limited to 1–1.5 μm to maintain low bending stiffness for minimized tissue damage. Both the interconnects and recording site surfaces were made of gold. A 20- μm diameter hole was designed at the tip of the flexible shank, in which the tip of the shuttling tungsten wire was anchored to drag the shank into brain tissue during implantation. The recording sites were coated with either 200 nm of sputtered iridium oxide film or electrochemically deposited PEDOT:PSS (poly(3,4-ethylenedioxythiophene) polystyrene sulfonate) to lower the electrode impedance to below 100 k Ω at 1 kHz in saline solution.

Each array was soldered to a 128-channel flexible printed circuit (FPC) board measuring 42 mm in length, which was connected to the SpikeGadgets 128-channel headstage (SpikeGadgets, San Francisco, USA) for signal acquisition. The four shuttling tungsten wires were fixed onto a carrier chip with 5% Poly (ethylene oxide)-300000 (PEO; CAS No. 25322-68-3) before implantation.

Surgery

Electrode array implantation

Electrode array implantation and viral injection for optogenetic inactivation experiments were performed before behavioral training. Mice were anesthetized with isoflurane before surgery (3–4% for induction, ~1% for maintenance) and head-fixed in a stereotaxic apparatus. Body temperature was maintained at 37°C using a heating pad. Chlortetracycline hydrochloride eye ointment was applied to prevent corneal drying. A circular piece of scalp was removed to expose the skull, and the incision site was treated with cyanoacrylate tissue adhesive (Vetbond, 3M, Saint Paul, USA).

For chronic implantation of uFINE-M arrays (Figure 1D), three craniotomies (~6 mm² each) were performed over the left hemisphere, and the dura was left intact. A grounding silver wire was implanted posterior to lambda on the right hemisphere. The cortical surface was kept moist with artificial cerebrospinal fluid or 1×phosphate-buffered saline (PBS). Arrays were implanted to OFC (orbitofrontal cortex, LO, VO and MO, AP 2.46 mm, ML 0.65 mm, depth 2.20 mm), across anterior M1 (primary motor cortex) and anterior M2 (secondary motor cortex, AP 1.94 mm, ML 1.50 mm, depth 0.80 mm), mPFC (medial prefrontal cortex, PL and IL, AP 1.78 mm, ML 0.30 mm, depth 2.20 mm), striatum (caudate putamen, AP 1.42 mm, ML 0.30 mm, depth 2.90 mm), across posterior M1, posterior M2, and ACC (anterior cingulate cortex, Cg1 and Cg2, AP -0.20 mm, ML 0.55 mm, depth 1.20 mm), MDTh (mediodorsal thalamus, MDL, MDC, MDM, AP -1.34 mm, ML 0.30 mm, deep 3.30 mm), across V1 (primary visual cortex) and V2L (secondary visual cortex lateral area, AP -2.80 mm, ML 3.25 mm, deep 0.90 mm), and V2M (secondary visual cortex medial area, V2MM and V2ML, AP -2.80 mm, ML 1.40 mm, deep 0.90 mm). Brain regions were named according to *The Mouse Brain in Stereotaxic Coordinates* by Franklin and Paxinos (3rd edition)³⁷. The shuttling tungsten wires were released from the carrier chip by applying saline to the 5% Poly (ethylene oxide)-300000 fixation site, and the tungsten wires were retracted from the brain 2 minutes after the implantation. The exposed parts of arrays were bonded together layer by layer using light-curable resin (Filtek™ Z350 XT, 3M, Saint Paul, USA). The craniotomy was sealed with a thin layer of silicone elastomer Kiwi-Cast (World Precision Instruments, Sarasota, USA). A custom-designed headplate was positioned on the skull and secured using Super-Bond C&B (SUN MEDICAL, Japan). After the Super-Bond Polymer cured, several layers of dental acrylic cement were applied to secure the entire implant.

Viral injection

For viral injections, the skull was not cracked but only thinned to allow smooth entry of a borosilicate glass pipette with a tip diameter of ~40–50 μm. A total of 150 nL viral solution—either AAV2/9-mCaMKIIa-eJaws 3.0-mRuby3-WPRE-pA (for manipulation group mice, TaiTool Bioscience, Shanghai, China) or AAV2/9-mCaMKIIa-mCherry-WPRE-pA (for control group mice, TaiTool Bioscience, Shanghai, China) were injected at a depth of 1750 μm for OFC and 500 μm for V2M using a syringe pump (Nanoject II Auto-Nanoliter Injector, Drummond Scientific Company, USA). Group identity (manipulation or control) was randomly assigned among cage mates. After injection, the pipette was left in place for 10–15 minutes before retraction. Optical fibers were implanted bilaterally above the virus injection sites (1000-μm deep for OFC and on cortical surface for V2M), angled ~10 degrees laterally. Mice were given carprofen (5 mg/kg) subcutaneously for postoperative analgesia.

Mice were allowed to recover from the surgery for at least 3 weeks before water-restriction and behavioral training.

Electrophysiological recording

Neural signals were amplified and recorded using a SpikeGadgets 1024-channel system (SpikeGadgets, San Francisco, USA). Raw voltage signals were sampled at 30 kHz. Task-related behavioral events were digitized as TTL signals and recorded simultaneously by the SpikeGadgets system.

Optogenetic inactivation

Optical silencing via activation of eJaws 3.0 activation was induced by LED red laser (625 nm; Thorlabs) and controlled by digital I/O devices (Arduino Uno R3, Arduino). To manipulate neural activity in either the OFC or V2M, the laser was delivered during stimulus period (0–800 ms after visual stimulus onset) or response period (800–2500 ms after visual stimulus onset) of all trials in separate manipulation groups. The laser power at the fiber tip was calibrated to 2 mW.

Histology

Mice were deeply anesthetized with isoflurane followed by an intraperitoneal injection of 15% ethyl carbamate solution. Transcardial perfusion was then performed using 4% paraformaldehyde (PFA). Brains were extracted, post-fixed in 4% PFA at 4°C overnight, and then transferred to 30% sucrose in PBS until equilibration for cryoprotection. Brains were sectioned at a thickness of 25 µm, and slices were mounted with antifade mounting medium (with DAPI). Fluorescence images were acquired using a virtual slide microscope (VS120, Olympus, Shinjuku, Japan; [Figure 8](#) and [Figure S1](#)).

Analysis of behavioral performance

To classify behavioral trial by task performance, we used the d-prime (d') metric²³ and labeled each trial by the d-prime value ([Figure S2](#)) calculated with the 10 trials before and 10 trials after it:

$$d' = \text{norminv}(\text{Hit rate}) - \text{norminv}(\text{FA rate}),$$

where *norminv* is the inverse of cumulative normal function, *Hit rate* is the frequency of Hit response in the Go stimulus trials, and *FA rate* is the frequency of False Alarm response in the No-Go stimulus trials. To minimize confounding effects of animal's motivation on the evaluation of task performance, the trials after the last lick in daily sessions were discarded ([Figure S2](#)).

Mouse oral-facial movements during training were recorded at 250 frames per second (fps) using a high-speed camera (MV-CA016-10UC, Hikrobot Co., Ltd., China). The video data were processed with the open-source software Facemap⁴⁵ and custom MATLAB scripts. ROIs were manually defined and the motion energy ([Figure S6](#)) at each timepoint was calculated as the absolute value of the difference between consecutive frames, summed across all pixels within ROI²⁸. To account for motion energy changes caused by environment luminance changes (e.g., LED response signal), we subtracted motion energy data of miss trials from the data of other trial types during the response period, as mice did not show observable oral-facial motion in miss trials.

Analysis of neuronal responses

Spike sorting

Spike sorting was performed offline using custom MATLAB scripts and open-source software Spyking Circus⁴⁶. Raw voltage signals were first filtered above 300 Hz, then denoised with manual threshold and common-median referenced within each probe shank⁴⁷. The preprocessed signals were fed to Spyking Circus, which applies automated density-based clustering and template-matching algorithm for spike detection. Spike detection threshold of each recording site was defined as 6 times of the median absolute deviation of the signal. Spatial whitening was then performed to remove spurious spatial correlation between nearby recording sites, and the spike detection thresholds were recalculated after whitening. The putative spike waveforms were aligned on local minima, projected to 5-dimensional feature space with principal component analysis, and the spike templates (putative cells) were constructed with density-based clustering. The temporal width of the spike templates was set as 3ms, the spatial width was set as 200 µm. The templates were finally matched to the data with an iterative greedy approach, with an assumption that templates sum linearly to find the spike times. The results from Spyking Circus were manually curated using Phy (<https://github.com/cortex-lab/phy>) to remove obvious artifacts with abnormal

waveform shape, and merge similar spike clusters in feature space. Spike clusters were considered as single units if the inter-spike interval exceeded 1 ms and the clusters were well isolated in feature space.

Firing rate and activity onset timing

To calculate the average firing rate, the spikes were first binned at 1-ms resolution and resolution spike rate was computed over a 25-ms time window. To account the higher noise in CR trials due to low trial number in some sessions, we constructed bootstrap-resampled datasets for 500 times, with only 5 trials sampled with replacement for each session in both the early stage and the expert stage to test if the low number of trials affect the results (Figure S3 [↗](#)).

To identify the activity onset timing of individual single units, the firing rate of a single unit in each 25-ms time window across trials was compared to its baseline activity (500–0 ms before the visual stimulus onset) by *t*-test. We identified time windows in which P values were below 0.05 for at least three consecutive time windows, and defined the first time window as the timing of activity onset (Figure 2 [↗](#)).

For each brain region, the time window with the highest proportion of neurons exhibiting activity onset was defined as the regional peak activation time, and the pairwise differences in peak activation times across all brain regions were used as a measure of the temporal compression of activation sequence (Figure S4 [↗](#)).

Connection rank analyses

Functional connections between neurons were defined based on the significance of cross-correlation scores between their spike trains. For each 200-ms timebin, cross-correlation scores between neurons were calculated with the total spiking probability edge (TSPE) algorithm ²⁶, in which an edge filter was applied to the cross-correlogram to facilitate the detection of local maxima and minima. A functional connection was identified if its cross-correlation score exceeded at least 95% of cross-correlation scores calculated from randomly shuffled spike trains. Only excitatory connections within 20 ms were included in subsequent analyses.

To establish regional connection profiles and identify key brain regions within the network, we defined the functional input/output strength between any two brain regions as the proportion of neuron pairs that had significant excitatory functional connections, of all possible input/output pairs between these two regions. To better evaluate the relative importance of each region within the brain network, we ranked the summed values of input/output strength of each brain region on a scale from 1 to 10 (Figures 3 [↗](#)-5 [↗](#), Figure 7 [↗](#), Figure S5, S7, S9 [↗](#)). To better compare interregional input/output strength, for each time window within a trial, regional connection strength was ranked on a scale of 1 to 10, with a rank of 1 representing the lowest 10% strength among all regional connections within the same time window (Figures 6 [↗](#), Figure S10 [↗](#)).

ROC analysis

To quantify the selectivity of each neuron for visual stimulus, we applied the receiver operating characteristic (ROC) analysis ²³ to the distributions of spike counts on each 200-ms time window within the trial. A neuron was included in the ROC analysis only if it had at least 5 trials for each of the four trial types (Hit, CR, Miss, and FA). For each neuron, 500 bootstrap-resampled data were generated in each time window, and the number of trials with different choices and visual stimuli were balanced to ensure 50 trials for each condition. The area under the ROC curve (auROC) indicates the accuracy with which an ideal observer can correctly classify whether a given response is recorded in one of the two conditions. ROC selectivity was defined as $2 \times \text{abs}(\text{auROC} - 0.5)$, which ranges from 0 to 1. A neuron was classified as stimulus-selective if its ROC selectivity was larger than 95% percent of randomly shuffled data ($p < 0.05$) in at least 95% of bootstrap-resampled dataset (Figure 7 [↗](#)).

Statistical analysis

No statistical methods were used to pre-determine sample sizes. Sample sizes were consistent with similar studies in the field. Statistical analyses were performed using MATLAB or GraphPad Prism (GraphPad Software). The two-way ANOVA were used to determine the significance of the effects. Correlation values were computed using Pearson's correlation. Unless otherwise specified, data were reported as mean $\pm$ SEM and statistical significance was set at $p < 0.05$.

Data availability

Spiking data and behavior data analyzed during this study and scripts of main analyses are available on Dryad: <https://doi.org/10.5061/dryad.cnp5hqcj2> .

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Additional information

Ethics

All experimental procedures were approved by the Animal Care and Use Committee at the Center for Excellence in Brain Science and Intelligence Technology, Chinese Academy of Sciences, protocol # NA-056-2020.

Author Contributions

Tian-Yi Wang and Chengcong Feng contributed equally to this work. Tian-Yi Wang, conceptualization, performed all the analyses on electrophysiology data and optogenetic manipulation, all the related behavior training experiments, drafting and revising the article; Chengcong Feng, performed all the implantation surgeries, video analyses of mouse oral-facial movements during task learning and related behavior training, design of ultra-flexible microelectrode array devices; Chengyao Wang, design and fabrication of ultra-flexible microelectrode array devices; Chi Ren, supervision, funding acquisition, revising the article; Zhengtuo Zhao, conceptualization, supervision, funding acquisition, revising the article.

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Additional files

[Supplemental Figures](#) 

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Peer reviews

Reviewer #1 (Public review):

Summary:

This study aims to address an important and timely question: how does the mesoscale architecture of cortical and subcortical circuits reorganize during sensorimotor learning? By using high-density, chronically implanted ultra-flexible electrode arrays, the authors track

spiking activity across ten brain regions as mice learn a visual Go/No-Go task. The results indicate that learning leads to more sequential and temporally compressed patterns of activity during correct rejection trials, alongside changes in functional connectivity ranks that reflect shifts in the relative influence of visual, frontal, and motor areas throughout learning. The emergence of a more task-focused subnetwork is accompanied by broader and faster propagation of stimulus information across recorded regions.

Strengths:

A clear strength of this work is its recording approach. The combination of stable, high-throughput multi-region recordings over extended periods represents a significant advance for capturing learning-related network dynamics at the mesoscale. The conceptual framework is well motivated, building on prior evidence that decision-relevant signals are widely distributed across the brain. The analysis approach, combining functional connectivity rankings with information encoding metrics is well motivated but needs refinement. These results provide some valuable evidence of how learning can refine both the temporal precision and the structure of interregional communication, offering new insights into circuit reconfiguration during learning.

Weaknesses:

Several important aspects of the evidence remain incomplete. In particular, it is unclear whether the reported changes in connectivity truly capture causal influences, as the rank metrics remain correlational and show discrepancies with the manipulation results. The absolute response onset latencies also appear slow for sensory-guided behavior in mice, and it is not clear whether this reflects the method used to define onset timing or factors such as task structure or internal state. Furthermore, the small number of animals, combined with extensive repeated measures, raises questions about statistical independence and how multiple comparisons were controlled. The optogenetic experiments, while intended to test the functional relevance of rank-increasing regions, leave it unclear how effectively the targeted circuits were silenced. Without direct evidence of reliable local inhibition, the behavioral effects or lack thereof are difficult to interpret.

<https://doi.org/10.7554/eLife.108083.2.sa3>

Reviewer #2 (Public review):

Summary:

Wang et al. measure from 10 cortical and subcortical brain as mice learn a go/no-go visual discrimination task. They found that during learning, there is a reshaping of inter-areal connections, in which a visual-frontal subnetwork emerges as mice gain expertise. Also visual stimuli decoding became more widespread post-learning. They also perform silencing experiments and find that OFC and V2M are important for the learning process. The conclusion is that learning evoked a brain-wide dynamic interplay between different brain areas that together may promote learning.

Strengths:

The manuscript is written well and the logic is rather clear. I found the study interesting and of interest to the field. The recording method is innovative and requires exceptional skills to perform. The outcomes of the study are significant, highlighting that learning evokes a widespread and dynamics modulation between different brain areas, in which specific task-related subnetworks emerge.

Weaknesses:

I had some major concerns that make the claims of the study less convincing: Low number of mice, insufficient movement analysis, figure visualization and analytic methods.

Nevertheless, I had several major concerns:

(1) The number of mice was small for the ephys recordings. Although the authors start with 7 mice in Figure 1, they then reduce to 5 in panel F. And in their main analysis they minimize their analysis 6/7 sessions from 3 mice only. I couldn't find a rationale for this reduction, but in the methods they do mention that 2 mice were used for fruitless training, which I found no mention in the results. Moreover, in the early case all of the analysis is from 118 CR trials taken from 3 mice. In general, this is a rather low number of mice and trial numbers. I think it is quite essential to add more mice.

(2) Movement analysis was not sufficient. Mice learning a go/no-go task establish a movement strategy that is developed throughout learning and is also biased towards Hit trials. There is an analysis of movement in Fig. S4 but this is rather superficial. I was not even sure that the 3 mice in Figure S4 are the same 3 mice in the main figure. There should be also an analysis of movement as a function of time to see differences. Also for Hits and FAs. I give some more details below. In general, most of the results can be explained by the fact that as mice gain expertise, they move more (also in CR during specific times) which leads to more activation in frontal cortex and more coordination with visual areas. More needs to be done in terms of analysis, or at least a mention of this in the text.

(3) Most of the figures are over-detailed and it is hard to understand the take home message. Although the text is written succinctly and rather short, the figures are mostly overwhelming, especially figures 4-7. For example, Figure 4 presents 24 brain plots! For rank input and output rank during early and late stim and response periods, for early and expert and their difference. All in the same colormap. No significance shown at all. The Δ rank maps for all cases look essentially identical across conditions. The division into early and late time periods is not properly justified. But the main take home message is positive Δ rank in OFC, V2M, V1 and negative Δ rank in ThalMD and Str. In my opinion, one trio maps is enough, and the rest could be bumped to the Supp, if at all. In general, the figures in several cases do not convey the main take home messages.

(4) Analysis is sometimes not intuitive enough. For example, the rank analysis of input and output rank seemed a bit over complex. Figure 3 was hard to follow (although a lot of effort was made by the authors to make it clearer). Was there any difference between output and input analysis? Also time period seem sometimes redundant. Also, there are other network analysis that can be done which are a bit more intuitive. The use of rank within the 10 areas was not the most intuitive. Even a dimensionality reduction along with clustering can be used as an alternative. In my opinion, I don't think the authors should completely redo their analysis, but maybe mention the fact that other analyses exist.

Reviewer comments to the authors' revision:

Thank you for the extensive revision. Most of my concerns were answered and the manuscript is much improved. Still, there are some major issues that remain unconvincing:

(1) The number of learning mice is only 3 which is substantially low as compared to other studies in the field. Thus, statistics are across trials and session pooled from all mice. This is a big limitation in supporting the authors' claims

(2) There is no measurement of movement during the task. Since there are already several studies showing that movement has a strong effect on brain-wide dynamics, and since it is well known that mice change their body movement during learning (at least some mice) the authors cannot disentangle between learning-related and movement-related dynamics. This

issue is properly discussed in the paper and also partially addressed with a control group where movement was measured without neural recordings.

(3) The authors do not know exactly where they recorded from, with emphasis on subcortical areas. The authors partially address this in a separate cohort where they regenerate the reproducibility rate of penetration locations, but still this is not a complete address to this concern.

Given the issues above, I strongly recommend including additional mice with body movement measurement in the future. Great job and congratulations on this study!

<https://doi.org/10.7554/eLife.108083.2.sa2>

Reviewer #3 (Public review):

Summary:

In the manuscript " Dynamics of mesoscale brain network during decision-making learning revealed by chronic, large-scale single-unit recording", Wang et al investigated mesoscale network reorganization during visual stimulus discrimination learning in mice using chronic, large-scale single-unit recordings across 10 cortical/subcortical regions. During learning, mice improved task performance mainly by suppressing licking on no-go trials. The authors found that learning induced restructuring of functional connectivity, with visual (V1, V2M) and frontal (OFC, M2) regions forming a task-relevant subnetwork during the acquisition of correct No-Go (CR) trials. Learning also compressed sequential neural activation and broadened stimulus encoding across regions. In addition, a region's network connectivity rank correlated with its timing of peak visual stimulus encoding. Optogenetic inhibition of orbitofrontal cortex (OFC) and high order visual cortex (V2M) impaired learning, validating its role in learning. The work highlights how mesoscale networks underwent dynamic structuring during learning.

Strengths:

The use of ultra-flexible microelectrode arrays (uFINE-M) for chronic, large-scale recordings across 10 cortical/subcortical regions in behaving mice represents a significant methodological advancement. The ability to track individual units over weeks across multiple brain areas will provide a rare opportunity to study mesoscale network plasticity.

While limited in scope, optogenetic inhibition of OFC and V2M directly ties connectivity rank changes to behavioral performance, adding causal depth to correlational observations.

Weaknesses:

The weakness is also related to the strength provided by the method. While the method in principle enables chronic tracking of individual units, the authors have not showed chronically tracked neurons across learning. Without demonstrating that and taking advantage of analyzing chronically tracked neurons, this approach is not different from acute recording in individual days across learning, weakening the attractiveness of the methodology and this study.

Another weakness is that major results are based on analyses of functional connectivity. Functional connection strengthen across areas is ranked 1-10 based on relative strength. And the regional input/out is compared across learning. This approach reveals differential changes in some cortical and subcortical areas. In my view, learning-related changes should be validated using complementary methods.

<https://doi.org/10.7554/eLife.108083.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Weaknesses:

The technical approach is strong and the conceptual framing is compelling, but several aspects of the evidence remain incomplete. In particular, it is unclear whether the reported changes in connectivity truly capture causal influences, as the rank metrics remain correlational and show discrepancies with the manipulation results.

We agree that our functional connectivity ranking analyses cannot establish causal influences. As discussed in the manuscript, besides learning-related activity changes, the functional connectivity may also be influenced by neuromodulatory systems and internal state fluctuations. In addition, the spatial scope of our recordings is still limited compared to the full network implicated in visual discrimination learning, which may bias the ranking estimates. In future, we aim to achieve broader region coverage and integrate multiple complementary analyses to address the causal contribution of each region.

The absolute response onset latencies also appear slow for sensory-guided behavior in mice, and it is not clear whether this reflects the method used to define onset timing or factors such as task structure or internal state.

We believe this may be primarily due to our conservative definition of onset timing. Specifically, we required the firing rate to exceed baseline (t-test, $p < 0.05$) for at least 3 consecutive 25-ms time windows. This might lead to later estimates than other studies, such as using the latency to the first spike after visual stimulus onset (Siegle et al., 2021) or the time to half-max response (Goldbach, Akitake, Leedy, & Histed, 2021).

The estimation of response onset latency in our study may also be affected by potential internal state fluctuations of the mice. We used the time before visual stimulus onset as baseline firing, since firing rates in this period could be affected by trial history, we acknowledge this may increase the variability of the baseline, thus increase the difficulty to statistically detect the onset of response.

Still, we believe these concerns do not affect the observation of the formation of compressed activity sequence in CR trials during learning.

Furthermore, the small number of animals, combined with extensive repeated measures, raises questions about statistical independence and how multiple comparisons were controlled.

We agree that a larger sample size would strengthen the robustness of the findings. However, as noted above, the current dataset has inherent limitations in both the number of recorded regions and the behavioral paradigm. Given the considerable effort required to achieve sufficient unit yields across all targeted regions, we wish to adjust the set of recorded regions, improve behavioral task design, and implement better analyses in future studies. This will allow us to both increase the number of animals and extract more precise insights into mesoscale dynamics during learning.

The optogenetic experiments, while intended to test the functional relevance of rank increasing regions, leave it unclear how effectively the targeted circuits were silenced.

Without direct evidence of reliable local inhibition, the behavioral effects or lack thereof are difficult to interpret.

We appreciate this important point. Due to the design of the flexible electrodes and the implantation procedure, bilateral co-implantation of both electrodes and optical fibers was challenging, which prevented us from directly validating the inhibition effect in the same animals used for behavior. In hindsight, we could have conducted parallel validations using conventional electrodes, and we will incorporate such controls in future work to provide direct evidence of manipulation efficacy.

Details on spike sorting are limited.

We have provided more details on spike sorting in method section, including the exact parameters used in the automated sorting algorithm and the subsequent manual curation criteria.

Reviewer #2 (Public review):

Weaknesses:

I had several major concerns:

(1) The number of mice was small for the ephys recordings. Although the authors start with 7 mice in Figure 1, they then reduce to 5 in panel F. And in their main analysis, they minimize their analysis to 6/7 sessions from 3 mice only. I couldn't find a rationale for this reduction, but in the methods they do mention that 2 mice were used for fruitless training, which I found no mention in the results. Moreover, in the early case, all of the analysis is from 118 CR trials taken from 3 mice. In general, this is a rather low number of mice and trial numbers. I think it is quite essential to add more mice.

We apologize for the confusion. As described in the Methods section, 7 mice (Figure 1B) were used for behavioral training without electrode array or optical fiber implants to establish learning curves, and an additional 5 mice underwent electrophysiological recordings (3 for visual-based decision-making learning and 2 for fruitless learning).

As we noted in our response to Reviewer #1, the current dataset has inherent limitations in both the number of recorded regions and the behavioral paradigm. Given the considerable effort required to achieve high-quality unit yields across all targeted regions, we wish to adjust the set of recorded regions, improve behavioral task design, and implement better analyses in future studies. These improvements will enable us to collect data from a larger sample size and extract more precise insights into mesoscale dynamics during learning.

(2) Movement analysis was not sufficient. Mice learning a go/no-go task establish a movement strategy that is developed throughout learning and is also biased towards Hit trials. There is an analysis of movement in Figure S4, but this is rather superficial. I was not even sure that the 3 mice in Figure S4 are the same 3 mice in the main figure. There should be also an analysis of movement as a function of time to see differences. Also for Hits and FAs. I give some more details below. In general, most of the results can be explained by the fact that as mice gain expertise, they move more (also in CR during specific times) which leads to more activation in frontal cortex and more coordination with visual areas. More needs to be done in terms of analysis, or at least a mention of this in the text.

Due to the limitation in the experimental design and implementation, movement tracking was not performed during the electrophysiological recordings, and the 3 mice shown in Figure S4 (now S5) were from a separate group. We have carefully examined the temporal profiles of mouse movements and found it did not fully match the rank dynamics for all

regions, and we have added these results and related discussion in the revised manuscript. However, we acknowledge the observed motion energy pattern could explain some of the functional connection dynamics, such as the decrease in face and pupil motion energy could explain the reduction in ranks for striatum.

Without synchronized movement recordings in the main dataset, we cannot fully disentangle movement-related neural activity from task-related signals. We have made this limitation explicit in the revised manuscript and discuss it as a potential confound, along with possible approaches to address it in future work.

(3) Most of the figures are over-detailed, and it is hard to understand the take-home message. Although the text is written succinctly and rather short, the figures are mostly overwhelming, especially Figures 4-7. For example, Figure 4 presents 24 brain plots! For rank input and output rank during early and late stim and response periods, for early and expert and their difference. All in the same colormap. No significance shown at all. The Δ rank maps for all cases look essentially identical across conditions. The division into early and late time periods is not properly justified. But the main take home message is positive Δ rank in OFC, V2M, V1 and negative Δ rank in ThalMD and Str. In my opinion, one trio map is enough, and the rest could be bumped to the Supplementary section, if at all. In general, the figure in several cases do not convey the main take home messages. See more details below.

We thank the reviewer for this valuable critique. The statistical significance corresponding to the brain plots (Figure 4 and Figure 5) was presented in Figure S3 and S5 (now Figure S5 and S7 in the revised manuscript), but we agree that the figure can be simplified to focus on the key results.

In the revised manuscript, we have condensed these figures to focus on the most important comparisons to make the visual presentation more concise and the take-home message clearer.

(4) The analysis is sometimes not intuitive enough. For example, the rank analysis of input and output rank seemed a bit over complex. Figure 3 was hard to follow (although a lot of effort was made by the authors to make it clearer). Was there any difference between the output and input analysis? Also, the time period seems redundant sometimes. Also, there are other network analysis that can be done which are a bit more intuitive. The use of rank within the 10 areas was not the most intuitive. Even a dimensionality reduction along with clustering can be used as an alternative. In my opinion, I don't think the authors should completely redo their analysis, but maybe mention the fact that other analyses exist

We appreciate the reviewer's comment. In brief, the input- and output-rank analyses yielded largely similar patterns across regions in CR trials, although some differences were observed in certain areas (e.g., striatum) in Hit trials, where the magnitude of rank change was not identical between input and output measures. We have condensed the figures to only show averaged rank results, and the colormap was updated to better convey the message.

We did explore dimensionality reduction applied to the ranking data. However, the results were not intuitive as well and required additional interpretation, which did not bring more insights. Still, we acknowledge that other analysis approaches might provide complementary insights.

Reviewer #3 (Public review):

Weaknesses:

The weakness is also related to the strength provided by the method. It is demonstrated in the original method that this approach in principle can track individual units for four months (Luan et al, 2017). The authors have not showed chronically tracked neurons across learning. Without demonstrating that and taking advantage of analyzing chronically tracked neurons, this approach is not different from acute recording across multiple days during learning. Many studies have achieved acute recording across learning using similar tasks. These studies have recorded units from a few brain areas or even across brain-wide areas.

We appreciate the reviewer's important point. We did attempt to track the same neurons across learning in this project. However, due to the limited number of electrodes implanted in each brain region, the number of chronically tracked neurons in each region was insufficient to support statistically robust analyses. Concentrating probes in fewer regions would allow us to obtain enough units tracked across learning in future studies to fully exploit the advantages of this method.

Another weakness is that major results are based on analyses of functional connectivity that is calculated using the cross-correlation score of spiking activity (TSPE algorithm). Functional connection strengthen across areas is then ranked 1-10 based on relative strength. Without ground truth data, it is hard to judge the underlying caveats. I'd strongly advise the authors to use complementary methods to verify the functional connectivity and to evaluate the mesoscale change in subnetworks. Perhaps the authors can use one key information of anatomy, i.e. the cortex projects to the striatum, while the striatum does not directly affect other brain structures recorded in this manuscript

We agree that the functional connectivity measured in this study relies on statistical correlations rather than direct anatomical connections. We plan to test the functional connection data with shorter cross-correlation delay criteria to see whether the results are consistent with anatomical connections and whether the original findings still hold.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) The small number of mice, each contributing many sessions, complicates the interpretation of the data. It is unclear how statistical analyses accounted for the small sample size, repeated measures, and non-independence across sessions, or whether multiple comparisons were adequately controlled.

We realized the limitation from the small number of animal subjects, yet the difficulty to achieve sufficient unit yields across all regions in the same animal restricted our sample size. Though we agree that a larger sample size would strengthen the robustness of the findings, however, as noted below the current dataset has inherent limitations in both the scope of recorded regions and the behavioral paradigm.

Given the considerable effort required to achieve sufficient unit yields across all targeted regions, we wish to adjust the set of recorded regions, improve behavioral task design, and implement better analyses in future studies. This will allow us to both increase the number of animals and extract more precise insights into mesoscale dynamics during learning.

(2) The ranking approach, although intuitive for visualizing relative changes in connectivity, is fundamentally descriptive and does not reflect the magnitude or reliability of the connections. Converting raw measures into ordinal ranks may obscure meaningful differences in strength and can inflate apparent effects when the underlying signal is weak.

We agree with this important point. As stated in the manuscript, our motivation in taking the ranking approach was that the differences in firing rates might bias cross-correlation between spike trains, making raw accounts of significant neuron pairs difficult to compare across conditions, but we acknowledge the ranking measures might obscure meaningful differences or inflate weak effects in the data.

We added the limitations of ranking approach in the discussion section and emphasized the necessity in future studies for better analysis approaches that could provide more accurate assessment of functional connection dynamics without bias from firing rates.

(3) The absolute response onset latencies also appear quite slow for sensory-guided behavior in mice, and it remains unclear whether this reflects the method used to determine onset timing or factors such as task design, sensorimotor demands, or internal state. The approach for estimating onset latency by comparing firing rates in short windows to baseline using a t-test raises concerns about robustness, as it may be sensitive to trial-to-trial variability and yield spurious detections.

We agree this may be primarily due to our conservative definition of onset timing. Specifically, we required the firing rate to exceed baseline (t-test, $p < 0.05$) for at least 3 consecutive 25-ms time windows. This might lead to later estimates than other studies, such as using the latency to the first spike after visual stimulus onset (Siegle et al., 2021) or the time to half-max response (Goldbach, Akitake, Leedy, & Histed, 2021).

The estimation of response onset latency in our study may also be affected by potential internal state fluctuations of the mice. We used the time before visual stimulus onset as baseline firing, since firing rates in this period could be affected by trial history, we acknowledge this may increase the variability of the baseline, thus increase the difficulty to statistically detect the onset of response.

Still, we believe these concerns do not affect the observation of the formation of compressed activity sequence in CR trials during learning.

(4) Details on spike sorting are very limited. For example, defining single units only by an interspike interval threshold above one millisecond may not sufficiently rule out contamination or overlapping clusters. How exactly were neurons tracked across days (Figure 7B)?

We have added more details on spike sorting, including the processing steps and important parameters used in the automated sorting algorithm. Only the clusters well isolated in feature space were accepted in manual curation.

We attempted to track the same neurons across learning in this project. However, due to the limited number of electrodes implanted in each brain region, the number of chronically tracked neurons in each region was insufficient to support statistically robust analyses.

This is now stated more clearly in the discussion section.

(5) The optogenetic experiments, while designed to test the functional relevance of rank-increasing regions, also raise questions. The physiological impact of the inhibition is not characterized, making it unclear how effectively the targeted circuits were actually silenced. Without clearer evidence that the manipulations reliably altered local activity, the interpretation of the observed or absent behavioral effects remains uncertain.

We appreciate this important point. Due to the design of the flexible electrodes and the implantation procedure, bilateral co-implantation of both electrodes and optical fibers was challenging, which prevented us from directly validating the inhibition effect in the same animals used for behavior. In hindsight, we could have conducted parallel validations using

conventional electrodes, and we will incorporate such controls in future work to provide direct evidence of manipulation efficacy.

(6) The task itself is relatively simple, and the anatomical coverage does not include midbrain or cerebellar regions, limiting how broadly the findings can be generalized to more flexible or ethologically relevant forms of decision-making.

We appreciate this advice and have expanded the existing discussion to more explicitly state that the relatively simple task design and anatomical coverage might limit the generalizability of our findings.

(7) The abstract would benefit from more consistent use of tense, as the current mix of past and present can make the main findings harder to follow. In addition, terms like "mesoscale network," "subnetwork," and "functional motif" are used interchangeably in places; adopting clearer, consistent terminology would improve readability.

We have changed several verbs in abstract to past form, and we now adopted a more consistent terminology by substituting “functional motif” as “subnetwork”. We still feel the use of

“mesoscale network” and “subnetwork” could emphasize different aspects of the results according to the context, so these words are kept the same.

(8) The discussion could better acknowledge that the observed network changes may not reflect task-specific learning alone but could also arise from broader shifts in arousal, attention, or motivation over repeated sessions.

We have expanded the existing discussion to better acknowledge the possible effects from broader shifts in arousal, attention, or motivation over repeated sessions.

(9) The figures would also benefit from clearer presentation, as several are dense and not straightforward to interpret. For example, Figure S8 could be organized more clearly to highlight the key comparisons and main message

We have simplified the over-detailed brain plots in Figure 4-5, and the plots in Figure 6 and S8 (now S10 in the revised manuscript).

(10) Finally, while the manuscript notes that data and code are available upon request, it would strengthen the study's transparency and reproducibility to provide open access through a public repository, in line with best practices in the field.

The spiking data, behavior data and codes for the core analyses in the manuscript are now shared in public repository (Dryad). And we have changed the description in the Data Availability section accordingly.

Reviewer #2 (Recommendations for the authors):

(A) Introduction:

(1) "Previous studies have implicated multiple cortical and subcortical regions in visual task learning and decision-making". No references here, and also in the next sentence.

The references were in the following introduction and we have added those references here as well.

We also added one review on cortical-subcortical neural correlates in goal-directed behavior (Cruz et al., 2023).

(2) Intro: In general, the citation of previous literature is rather minimal, too minimal. There is a lot of studies using large scale recordings during learning, not necessarily visual tasks. An example for brain-wide learning study in subcortical areas is Sych et al. 2022 (cell reports). And for wide-field imaging there are several papers from the Helmchen lab and Komiyama labs, also for multi-area cortical imaging.

We appreciate this advice. We included mainly visual task learning literature to keep a more focused scope around the regions and task we actually explored in this study. We fear if we expand the intro to include all the large-scale imaging/recording studies in learning field, the background part might become too broad.

We have included (Sych, Fomins, Novelli, & Helmchen, 2022) for its relevance and importance in the field.

(3) In the intro, there is only a mention of a recording of 10 brain regions, with no mention of which areas, along with their relevance to learning. This is mentioned in the results, but it will be good in the intro.

The area names are now added in intro.

(B) Results:

(1) Were you able to track the same neurons across the learning profile? This is not stated clearly.

We did attempt to track the same neurons across learning in this project. However, due to the limited number of electrodes implanted in each brain region, the number of chronically tracked neurons in each region was insufficient to support statistically robust analyses.

We now stated this more clearly in the discussion section.

(2) Figure 1 starts with 7 mice, but only 5 mice are in the last panel. Later it goes down to 3 mice. This should be explained in the results and justified.

We apologize for the confusion. As described in the Methods section, 7 mice (Figure 1B) were used for behavioral training without electrode array or optical fiber implants to establish learning curves, and an additional 5 mice underwent electrophysiological recordings (3 for visual-based decision-making learning and 2 for fruitless learning).

(3) I can't see the electrode tracks in Figure 1d. If they are flexible, how can you make sure they did not bend during insertion? I couldn't find a description of this in the methods also.

The electrode shanks were ultra-thin (1-1.5 μm) and it was usually difficult to recover observable tracks or electrodes in section.

The ultra-flexible probes could not penetrate brain on their own (since they are flexible), and had to be shuttled to position by tungsten wires through holes designed at the tip of array shanks. The tungsten wires were assembled to the electrode array before implantation; this was described in the section of electrode array fabrication and assembly. We also included the description about the retraction of the guiding tungsten wires in the surgery section to avoid confusion.

As an further attempt to verify the accuracy of implantation depth, we also measured the repeatability of implantation in a group of mice and found a tendency for the arrays to end in slightly deeper location in cortex ($142.1 \pm 55.2 \mu\text{m}$, $n = 7$ shanks), and slightly shallower location in subcortical structure ($-122.6 \pm 71.7 \mu\text{m}$, $n = 7$ shanks). We added these results as new Figure S1 to accompany Figure 1.

(4) In the spike raster in 1E, there seems to be ~20 cells in V2L, for example, but in 1F, the number of neurons doesn't go below 40. What is the difference here?

We checked Figure 1F, the plotted dots do go below 40 to ~20. Perhaps the file that reviewer received wasn't showing correctly?

(5) The authors focus mainly on CR, but during learning, the number of CR trials is rather low (because they are not experts). This can also be seen in the noisier traces in Figure 2a. Do the authors account for that (for example by taking equal trials from each group)?

We accounted this by reconstructing bootstrap-resampled datasets with only 5 trials for each session in both the early stage and the expert stage. The mean trace of the 500 datasets again showed overall decrease in CR trial firing rate during task learning, with highly similar temporal dynamics to the original data.

The figure is now added to supplementary materials (as Figure S3 in the revised manuscript).

(6) From Figure 2a, it is evident that Hit trials increase response when mice become experts in all brain areas. The authors have decided to focus on the response onset differences in CRs, but the Hit responses display a strong difference between naïve and expert cases.

Judged from the learning curve in this task the mice learned to inhibit its licking action when the No-Go stimuli appeared, which is the main reason we focused on these types of trials.

The movement effects and potential licking artefacts in Hit trials also restricted our interpretation of these trials.

(7) Figure 3 is still a bit cumbersome. I wasn't 100% convinced of why there is a need to rank the connection matrix. I mean when you convert to rank, essentially there could be a meaningful general reduction in correlation, for example during licking, and this will be invisible in the ranking system. Maybe show in the supp non-ranked data, or clarify this somehow

We agree with this important point. As stated in the manuscript and response to Reviewer #1, our motivation in taking the ranking approach was that the differences in firing rates could bias cross-correlation between spike trains, making raw accounts of significant neuron pairs difficult to compare across conditions, but we acknowledge the ranking measures might obscure meaningful differences or inflate weak effects in the data.

We added the limitations of ranking approach in the discussion section and emphasized the necessity in future studies for better analysis approaches that could provide more accurate assessment of functional connection dynamics without bias from firing rates.

(8) Figure 4a x label is in manuscript, which is different than previous time labels, which were seconds.

We now changed all time labels from Figure 2 to milliseconds.

(9) Figure 4 input and output rank look essentially the same.

We have compressed the brain plots in Figures 4-5 to better convey the take-home message.

(10) Also, what is the late and early stim period? Can you mark each period in panel A? Early stim period is confusing with early CR period. Same for early response and late response.

The definition of time periods was in figure legends. We now mark each period out to avoid confusion.

(11) Looking at panel B, I don't see any differences between delta-rank in early stim, late stim, early response, and late response. Same for panel c and output plots.

The rankings were indeed relatively stable across time periods. The plots are now compressed and showed a mean rank value.

(12) Panels B and C are just overwhelming and hard to grasp. Colors are similar both to regular rank values and delta-rank. I don't see any differences between all conditions (in general). In the text, the authors report only M2 to have an increase in rank during the response period. Late or early response? The figure does not go well with the text. Consider minimizing this plot and moving stuff to supplementary.

The colormap are now changed to avoid confusion, and brain plots are now compressed.

(13) In terms of a statistical test for Figure 4, a two-way ANOVA was done, but over what? What are the statistics and p-values for the test? Is there a main effect of time also? Is there a significant interaction? Was this done on all mice together? How many mice? If I understand correctly, the post-hoc statistics are presented in the supplementary, but from the main figure, you cannot know what is significant and what is not.

For these figures we were mainly concerned with the post-hoc statistics which described the changes in the rankings of each region across learning.

We have changed the description to “t-test with Sidak correction” to avoid the confusion.

(14) In the legend of Figure 4, it is reported that 610 expert CR trials from 6 sessions, instead of 7 sessions. Why was that? Also, like the previous point, why only 3 mice?

Behavior data of all the sessions used were shown in Figure S1. There were only 3 mice used for the learning group, the difficulty to achieve sufficient unit yields across all regions in the same animal restricted our sample size

(15) Body movement analysis: was this done in a different cohort of mice? Only now do I understand why there was a division into early and late stim periods. In supp 4, there should be a trace of each body part in CR expert versus naïve. This should also be done for Hit trials as a sanity check. I am not sure that the brightness difference between consecutive frames is the best measure. Rather try to calculate frame-to-frame correlation. In general, body movement analysis is super important and should be carefully analyzed.

Due to the limitation in the experimental design and implementation, movement tracking was not performed during the electrophysiological recordings, and the 3 mice shown in Figure S4 (now S5) were from a separate group. We have carefully examined the temporal profiles of mouse movements and found it did not fully match the rank dynamics for all regions, and we have added these results and related discussion in the revised manuscript. However, we acknowledge the observed motion energy pattern could explain some of the functional connection dynamics, such as the decrease in face and pupil motion energy could explain the reduction in ranks for striatum.

Without synchronized movement recordings in the main dataset, we cannot fully disentangle movement-related neural activity from task-related signals. We have made this limitation explicit in the revised manuscript and discuss it as a potential confound, along with possible approaches to address it in future work.

(16) For Hit trials, in the striatum, there is an increase in input rank around the response period, and from Figure S6 it is clear that this is lick-related. Other than that, the authors report other significant changes across learning and point out to Figure 5b,c. I couldn't see which areas and when it occurred.

We did naturally expect the activity in striatum to be strongly related to movement.

With Figure S6 (now S7) we wished to show that the observed rank increase for striatum could not simply be attributed to changes in time of lick initiation.

As some readers may argue that during learning the mice might have learned to only intensely lick after response signal onset, causing the observed rise of input rank after response signal, we realigned the spikes in each trial to the time of the first lick, and a strong difference could still be observed between early training stage and expert training stage.

We still cannot fully rule out the effects from more subtle movement changes, as the face motion energy did increase in early response period. This result and related discussion has been added to the results section of revised manuscript.

(17) Figure 6, again, is rather hard to grasp. There are 16 panels, spread over 4 areas, input and output, stim and response. What is the take home message of all this? Visually, it's hard to differentiate between each panel. For me, it seems like all the panels indicate that for all 4 areas, both in output and input, frontal areas increase in rank. This take-home message can be visually conveyed in much less tedious ways. This simpler approach is actually conveyed better in the text than in the figures themselves. Also, the whole explanation on how this analysis was done, was not clear from the text. If I understand it, you just divided and ranked the general input (or output) into individual connections? If so, then this should be better explained.

We appreciate this advice and we have compressed the figures to better convey the main message. The rankings for Figure 6 and Figure S8 (now Figure S9) was explained in the left panel of Figure 3C. Each non-zero element in the connection matrix was ranked to value from 1-10, with a value of 10 represented the 10% strongest non-zero elements in the matrix.

We have updated the figure legends of Figure 3, and we have also updated the description in methods (Connection rank analyses) to give a clearer description of how the analyses were applied in subsequent figures.

(18) Figure 7: Here, the authors perform a ROC analysis between go and no-go stimuli. They balance between choice, but there is still an essential difference between a hit and a FA in terms of movement and licks. That is maybe why there is a big difference in selective units during the response period. For example, during a Hit trial the mouse licks and gets a reward, resulting in more licking and excitement. In FAs, the mouse licks, but gets punished, which causes a reduction in additional licking and movements. This could be a simple explanation why the ROC was good in the late response period. Body movement analysis of Hit and FA should be done as in Figure S4.

We appreciate this insightful advice.

Though we balanced the numbers of basic trial types, we couldn't rule out the difference in the intrinsic movement amount difference in FA trials and Hit trials, which is likely the reason of large proportion of encoding neurons in response period.

We have added this discussion both in result section and discussion section along with the necessity of more carefully designed behavior paradigm to disentangle task information.

(19) The authors also find selective neurons before stimulus onset, and refer to trial history effects. This can be directly checked, that is if neurons decode trial history.

We attempted encoding analyses on trial history, but regrettably for our dataset we could not find enough trials to construct a dataset with fully balanced trial history, visual stimulus and behavior choice.

(20) Figure 7e. What is the interpretation for these results? That areas which peaked earlier had more input and output with other areas? So, these areas are initiating hubs? Would be nice to see ACC vs Str traces from B superimposed on each other. Having said this, the Str is the only area to show significant differences in the early stim period. But is also has the latest peak time. This is a bit of a discrepancy.

We appreciate this important point.

The limitation in the anatomical coverage of brain regions restricted our interpretation about these findings. They could be initiating hubs or earlier receiver of the true initiating hubs that were not monitored in our study.

The Str trace was in fact above the ACC trace, especially in the response period. This could be explained by the above advice 18: since we couldn't rule out the difference in the intrinsic movement amount difference in FA trials and Hit trials, and considering striatum activity is strongly related to movement, the Str trace may reflect more in the motion related spike count difference between FA trials and Hit trials, instead of visual stimulus related difference.

This further shows the necessity of more carefully designed behavior paradigm to disentangle task information.

The striatum trace also in fact didn't show a true double peak form as traces in other regions, it ramped up in the stimulus region and only peaked in response period. This description is now added to the results section.

In the early stim period, the Striatum did show significant differences in average percent of encoding neurons, as the encoding neurons were stably high in expert stage. The striatum activity is more directly affected Still the percentage of neurons only reached peak in late stimulus period.

(21) For the optogenetic silencing experiments, how many mice were trained for each group? This is not mentioned in the results section but only in the legend of Figure 8. This part is rather convincing in terms of the necessity for OFC and V2M

We have included the mice numbers in results section as well.

(C) Discussion

(1) There are several studies linking sensory areas to frontal networks that should be mentioned, for example, Esmaili et al, 2022, Matteucci et al., 2022, Guo et al, 2014, Gallero Salas et al, 2021, Jerry Chen et al, 2015. Sonja Hofer papers, maybe. Probably more.

We appreciate this advice. We have now included one of the mentioned papers (Esmaili et al., 2022) in the results section and discussion section for its direct characterization of the enhanced coupling between somatosensory region and frontal (motor) region during sensory learning. The other studies mentioned here seem to focus more on the differences in encoding properties between regions along specific cortical pathways, rather than functional connection or interregional activity correlation, and we feel they are not directly related to the observations discussed.

(2) *The reposted reorganization of brain-wide networks with shifts in time is best described also in Sych et al. 2021.*

We regret we didn't include this important research and we have now cited this in discussion section.

(3) *Regarding the discussion about more widespread stimulus encoding after learning, the results indicate that the striatum emerges first in decoding abilities (Figure 7c left panel), but this is not discussed at all.*

We briefly discussed this in the result section. We tend to attribute this to trial history signal in striatum, but since the structure of our data could not support a direct encoding analysis on trial history, we felt it might be inappropriate to over-interpret the results.

(4) *An important issue which is not discussed is the contribution of movement which was shown to have a strong effect on brain-wide dynamics (Steinmetz et al 2019; Musall et al 2019; Stringer et al 2019; Gilad et al 2018) The authors do have some movement analysis, but this is not enough. At least a discussion of the possible effects of movement on learning-related dynamics should be added.*

We have included these studies in discussion section accordingly. Since the movement analyses were done in a separate cohort of mice, we have made our limitation explicit in the revised manuscript and discuss it as a potential confound, along with possible approaches to address it in future work.

(D) *Methods*

(1) *How was the light delivery of the optogenetic experiments done? Via fiber implantation in the OFC? And for V2M? If the red laser was on the skull, how did it get to the OFC?*

The fibers were placed on cortex surface for V2M group, and were implanted above OFC for OFC manipulation group. These were described in the viral injection part of the methods section.

(2) *No data given on how electrode tracking was done post hoc*

As noted in our response to the advice 3 in results section, the electrode shanks were ultra-thin (1-1.5 μm) and it was usually difficult to recover observable tracks or electrodes in section.

As an attempt to verify the accuracy of implantation depth, we measured the repeatability of implantation in a group of mice and found a tendency for the arrays to end in slightly deeper location in cortex ($142.1 \pm 55.2 \mu\text{m}$, $n = 7$ shanks), and slightly shallower location in subcortical structure ($-122.6 \pm 71.7 \mu\text{m}$, $n = 7$ shanks). We added these results as new Figure S1 to accompany Figure 1.

Reviewer #3 (Recommendations for the authors):

(1) *The manuscript uses decision-making in the title, abstract and introduction. However, nothing is related to decision learning in the results section. Mice simply learned to suppress licking in no-go trials. This type of task is typically used to study behavioral inhibition. And consistent with this, the authors mainly identified changes related to network on no-go trials. I really think the title and main message is misleading. It is better to rephrase it as visual discrimination learning. In the introduction, the authors also reviewed multiple related studies that are based on learning of visual discrimination tasks.*

We do view the Go/No-Go task as a specific genre of decision-making task, as there were literature that discussed this task as decision-making task under the framework of signal detection theory or updating of item values (Carandini & Churchland, 2013; Veling, Becker, Liu, Quandt, & Holland, 2022).

We do acknowledge the essential differences between the Go/No-Go task and the tasks that require the animal to choose between alternatives, and since we have now realized some readers may not accept this task as a decision task, we have changed the title to visual discrimination task as advised.

(2) Learning induced a faster onset on CR trials. As the no-go stimulus was not presented to mice during early stages of training, this change might reflect the perceptual learning of relevant visual stimulus after repeated presentation. This further confirms my speculation, and the decision-making used in the title is misleading.

We have changed the title to visual discrimination task accordingly.

(3) Figure 1E, show one hit trial. If the second 'no-go stimulus' is correct, that trial might be a false alarm trial as mice licked briefly. I'd like to see whether continuous licking can cause motion artifacts in recording.

We appreciate this important point. There were indeed licking artifacts with continuous licking in Hit trials, which was part of the reason we focused our analyses on CR trials. Opto-based lick detectors may help to reduce the artefacts in future studies.

(4) What is the rationale for using a threshold of $d' < 2$ as the early-stage data and $d' > 3$ as expert stage data?

The thresholds were chosen as a result from trade-off based on practical needs to gather enough CR trials in early training stage, while maintaining a relatively low performance.

Assume the mice showed lick response in 95% of Go stimulus trials, then $d' < 2$ corresponded to the performance level at which the mouse correctly rejected less than 63.9% of No-Go stimulus trials, and $d' > 3$ corresponded to the performance level at which the mouse correctly rejected more than 91.2% of No-Go stimulus trials.

(5) Figure 2A, there is a change in baseline firing rates in V2M, MDTh, and Str. There is no discussion. But what can cause this change? Recording instability, problem in spiking sorting, or learning?

It's highly possible that the firing rates before visual stimulus onset is affected by previous reward history and task engagement states of the mice. Notably, though recorded simultaneously in same sessions, the changes in CR trials baseline firing rates in the V2M region were not observed in Hit trials.

Thus, though we cannot completely rule out the possibility in recording instability, we see this as evidence of the effects on firing rates from changes in trial history or task engagement during learning.

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